Musings on Markets

My not-so-profound thoughts about valuation, corporate finance and the news of the day!

Thursday, June 4, 2026

Revisiting the SpaceX Valuation: A Post-Prospectus Update!

A few weeks ago, I assessed the value of SpaceX ahead of its initial public offering, with the admission that I was making my estimates with drabs of data, some of it coming from unofficial sources. I also promised to revisit my valuation, when the prospectus came out, and now that it has, I will examine how the information it contains has changed my view of the company and its valuation. I will also use this post to talk about the information gained by having access to a company's financials, and why the information you glean from those financials is different at younger companies, with growth potential, relative to mature companies.

The Prospectus: Data versus Information

The requirement that companies that plan to go public in the United States have to register with the Securities Exchange Commission (SEC) and file a prospectus has been in place for decades, but the contents have changed over time, with disclosures added on partly by regulation and partly in response to investor demands. In a paper focusing on IPO disclosures from a couple of years ago, I noted that prospectuses have become more bloated over time, often running four to five times longer than those filed by companies that went public three or four decades ago, but not necessarily more informative. The SpaceX prospectus that we made public on May 20, 2026, is 277 pages long, with an addendum that runs another 100 pages, with dozens of pictures (mostly of spaceships going into orbit), a soaring story, but with weak links and multiple distractions. To get a measure of how the prospectus changes my pre-prospectus story and valuation, I will start with the easy part of the update, where I use the numbers from the financial statements in the prospectus to replace my pre-prospectus estimates, on operating metrics like revenues and earnings as well as on share count and IPO proceeds. I will then move

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on to the weightier part of the analysis, where I assess how the information in the prospectus has changed my story line and value for the company.

The Prospectus: Data update

In my pre-prospectus valuation, where I assessed the value of SpaceX at roughly $1.2 trillion, I relied on scraps of information, including leaked stories of estimated revenues ($15.5 billion) and EBITDA of $8 billion, since I did not have access to the company's full financial statements. With the release of the prospectus, that shortcoming has been remedied, and I started by updating the operating metrics that drive the intrinsic value of the company:

	Pre-prospectus (my estimates $ millions)	*Prospectus ($ millions)*
Revenues (Launch)	$4,100.00	$4,086.00
Revenues (Starlink)	$11,400.00	$11,387.00
Revenues (xAI)	$100.00	$3,201.00
Operating income or EBIT	-$2,000.00	-$2,589.00
Interest expense	$ -	$1,945.00
Net Income	Not estimated	-$4,937.00
R&D Expenses	Not available	Capitalized over 5 years
Operating income (adjusted for R&D)	Not available	$4,020.00

SpaceX prospectus

As you can see, my estimates for revenues for the launch and connectivity (Starlink) businesses were close to the actual numbers, but my xAI revenue estimates were much lower than reported. Overall, I had estimated an operating loss of $2 billion in 2025, and the prospectus yielded a larger loss of $2.57 billion. With almost $2 billion in interest expenses, unavailable prior to the prospectus, incorporated, the company reported a net loss of about $5 billion. A big factor in the operating losses reported by the company were its ballooning R&D expenses, and in keeping with my argument that these expenses should be capitalized, I estimated an earnings before interest, taxes and R&D of $4 billion.

On the financing front, the prospectus filled in details on cash and debt that were unavailable prior to the prospectus being made public:

	Pre-prospectus (my estimates $ millions)	*Prospectus ($ millions)*
Book value of equity	$20,000.00	$41,325.00
Book value of debt	$0.00	$22,896.00
Cash and Marketable Securities	$0.00	$24,747.00
Net Debt	$0.00	-$1,851.00

SpaceX prospectus

My pre-prospectus estimate of book value of equity was a shot in the dark, at $20 billion, but the acquisition of xAI caused that number to jump to $41.3 billion, as did the total debt (inclusive of leases) to $22.9 billion. The former (book value of equity) played little role in my valuation, but ignoring debt of this magnitude may seem monumental, there are two offsetting factors that reduce the impact on my value estimate. The first is that I also ignored the presence of cash, and with $24.7 billion in cash, the company's net debt is −$1.9 billion (cash exceeds debt), making the impact on value minimal. The second is that with my estimate enterprise value of $1.21 trillion, the debt, even if considered in full, is small enough to represent rounding error.

The prospectus did contain information on share count and structure, as well as on the company's plans for the proceeds, and both were useful at the margin, with the former affecting

About Me



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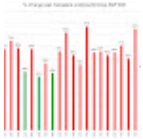
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my estimated value per share and the latter determining the treatment of the cash that will be raised from the offering:

- Share count: In my initial valuation, I used the private company pricing per share in conjunction with estimated market cap to back out a share count of 2467 million shares. With the prospectus, we get a clearer sense of shares outstanding, with a basic share count of 12,535 million shares reported in the prospectus (pages 246 & 247) in computing per share numbers. That share count does not include the new shares that will be issued in the offering, but that share count will be determined by the magnitude of the offering as well as the expected issuance price, and while the total share count includes options, warrants and rights that are exercisable before June 30, it does not include restricted stock units held by employees (see prospectus, page 18) and that information is still blanked out in the prospectus.
- Use of proceeds: It is estimated that SpaceX plans to raise $75 billion from the offering, and the prospectus specifies that the company plans to hold the proceeds to cover infrastructure investments in these businesses (see prospectus, page 66). That implies that any money raised in the offering will add to the company's cash balance, right after the offering, and will augment firm value (but not enterprise value).

The prospectus also lays bare the governance questions that will overhang the firm, with information that there will be two classes of shares- 6,932 million class A shares with one vote per share and 5,602 million class B shares with ten votes per share. The public offering will be class A shares, and with Elon Musk holding all of the class B shares, he will control more than 85% of the voting rights in the company. In summary, the prospectus is long and filled with distractions, but there is almost nothing in it that surprises me. SpaceX is a growing company that is money-losing and cash-burning, that will be a Elon Musk vehicle (with all the pluses and minuses that entails).

The Prospectus: Story update

In my original post, I noted that SpaceX is a company, where it is the story about how its businesses will evolve over time that drives value, rather than the base year numbers (on revenues, earnings and cash flows). That story, broadly speaking, has three key spokes to it and they are summarized below:



science, and I argue that it is neither; it does not hav...



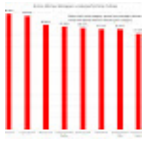
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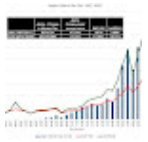
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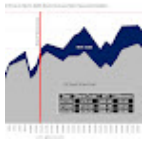
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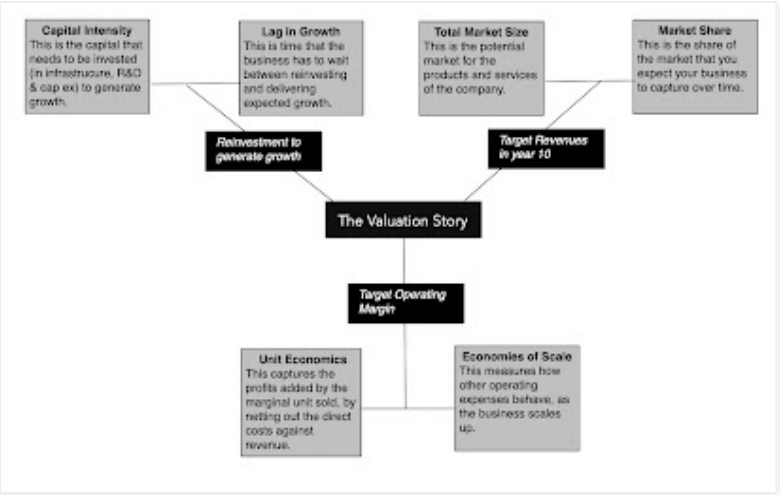
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The first of these spokes, *target revenues,* frame how big each business can grow over time, and is a function of the total market and market share. The second, the *target operating margin*, will capture how profitable each business can become, and is determined by unit economics and economies of scale. The third, reinvestment, measures how much each business has to invest to get to target revenues, and will vary with the capital intensity of the business. To frame how my valuation will change, as a result of what I learned from looking at the prospectus, I will start by presenting by pre-prospectus estimates on these key inputs, and then look at the impact of the prospectus on each input.

Pre-prospectus inputs and value

My pre-prospectus valuation of SpaceX contains my storyline for the three businesses that the company is in, with an add-on for the expansion options embedded in each business:

Business	Storyline	Revenues in 2036 ($ millions)	Operating Margin in 2036	Sales to Capital: 2026-30	Sales to Capital: 2031-2035
Launch	Dominant player in a growing launch market; 70% market share of $100 billion market.	$70,000	40%	4.00	2.00
Starlink	Market leader (75%) in satellite broadband, a niche (10%) of very large broadband service market ($1.5 trillion)	$120,000	60%	10.00	5.00
xAI	LLM with focus primarily on consumer subcriptions, rather than business-use	$80,000	45%	2.50	1.50
Expansion Options	Expansion options in space (travel), internet services and AI-related businesses.	$50,000	30%	3.00	3.00

With these inputs in place, I estimated a value of $1.2 trillion the SpaceX enterprise, and since I ignored cash and debt, this yielded an equivalent market value. Driving these numbers are upbeat stories about each of the three businesses that SpaceX is in, with large revenues and high margins in stable growth.



Data Update 8 for 2026: Time for Harvesting - Dividends and Buybacks

In the data update posts this year, I have wended my way from the macro (equities collectives, the bond market and other asset classes) ...

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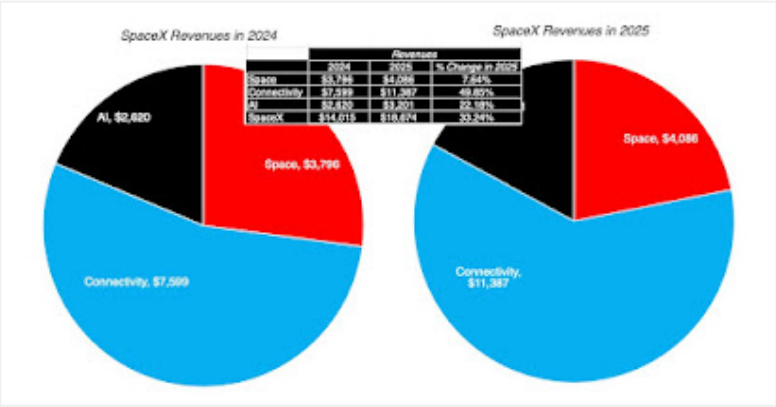
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► 2018 (35)
► 2017 (28)
► 2016 (48)
► 2015 (50)
► 2014 (44)

The Prospectus Effect

To the extent that the prospectus contains information that alters the storylines on any or all of these businesses, it will affect my estimate of value for SpaceX.

1. Revenue Growth (Target Revenues)

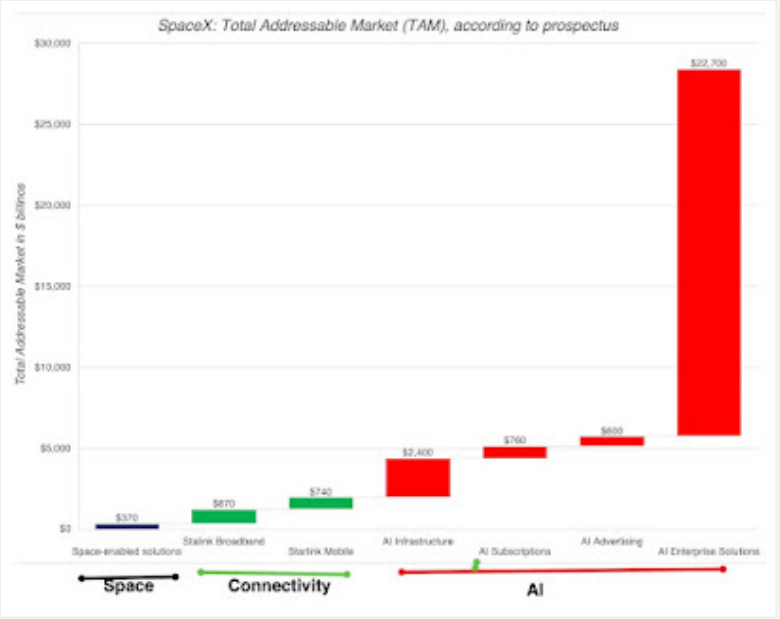
I will start with the growth (target revenues) input and use two parts of the prospectus to reexamine my story. The first is the *historical growth* reported by the company for each of its three business lines - launch, connectivity and AI.



As you can see, the company saw its revenues grow by a third in 2025, relative to 2024, with divergence across businesses; the connectivity business led with revenues growing by almost 50%, the AI business saw an increase in revenues of about 22% but the space business reported only modest growth in the year (7.64%). In short, notwithstanding the star role played by AI and the appeal of the rockets in the space launch business, it is Starlink that carried the company in 2025. The prospectus mentions Colossus, xAI's compute center, which has been leased to Anthropic for an eye-popping $1.25 billion a month, which should kickstart revenues next year, with the potential of tension in future years if xAI plans to go head-to-head against Anthropic in the AI products market.

The other relevant section of the prospectus contained estimates of total addressable market (TAM) for the company, broken down by business:

► 2013 (36)
► 2012 (49)
► 2011 (55)
► 2010 (45)
► 2009 (60)
► 2008 (42)



If the prospectus is to be believed, SpaceX has the largest TAM of any company in history, with a total TAM of $28 trillion, and AI accounts for $26 trillion of that market estimate. This estimate borders on fantasy, but I will cut the bankers who came up with these numbers some slack for two reasons. First, the estimation of TAM has been gamified by Silicon Valley, with bloated and patently unreachable numbers floated for companies, as I noted when I valued Uber (which was given a TAM of $5.7 trillion in its prospectus) for its IPO in 2019 and Airbnb (with a TAM of $3.4 trillion in its prospectus) in 2020. Second, it is true that AI agents are usable across almost every business and geography, giving it much wider reach than most products and services, and while the details of how the TAM was estimated are not specified in the prospectus, my guess is that the $26 trillion estimate includes all or most of the operating expenses of all businesses.

Story takeaway: I will stick with my estimates for target markets for the space launch and connectivity businesses, since the TAMs in the prospectus are, in my view, over reaches, and I will slow growth in the near years, to reflect that these businesses will take time to mature. In the AI business, I disagree with the magnitude of the TAM in the prospectus, but the acquisition of Cursor and the indications in the prospectus suggest that xAI very much wants to be part of the enterprise solutions space, notwithstanding its immense capitalization needs, and I will double my target revenues for AI from $80 billion to $160 billion, reflecting my estimate of a TAM of about $3 trillion to $4 trillion for AI products and services from businesses.

2. Profitability

On the profitability front, the first part of the prospectus that I looked at was its breakdown of income statements, by business:

Space Business
Has the best unit economics, and showed improvement year-t-year, reflecting the superior economics of reusable spacecraft in space launches. R&D takes a substantial bite, but its effect should level off, as growth steadies.

Connectivity Business
Solid and improving unit economics, notwithstanding lower revenues per subscriber. As with space business, R&D has significant effect on operating income, but here again, treating R&D as a capital expense will cause its effect to level off.

SpaceX: Unit Economics, by business

	Space		Connectivity		AI	
	2024	2025	2024	2025	2024	2025
Revenues	$3,796	$4,086	$7,599	$11,387	$2,620	$3,201
Cost of goods sold	1,541	$1,352	$4,768	$5,921	$1,687	$2,178
Gross Income	$2,255	$2,734	$2,831	$5,466	$933	$1,023
R&D Expenses	1,835	$3,004	$1,176	$575	$1,176	$5,064
Other oper expense	399	$387	$372	$468	$1,318	$1,827
Opeating Income	$21	-$657	$1,283	$4,423	-$1,561	-$5,868
Gross Margin	59.40%	66.91%	37.25%	48.00%	35.61%	31.96%
Operating Margin	0.55%	-16.08%	16.88%	38.84%	-59.58%	-183.32%
EBITR&D Margin	48.89%	57.44%	32.36%	43.89%	-14.69%	-25.12%

AI Business
Least attractive unit economics, with worsening in 2025, driven partly by the cost of delivering on AI subcriptions and services and partly be competitive pricing. R&D is substantial, and witht the capital expenditures in this business, it is the most capital intensive of the three businesses.

With the caveat that we have only two years of detailed information, there are interesting findings that emerge from the historical data on each of the businesses.

- *The space business has the best unit economics of the three business*, with a gross margin of about 67%, reflecting the cost advantages of its reusable rocket technology. While the space business reported an operating loss, that was entirely because of its weighty R&D expenses, and capitalizing those expenses results in a healthy operating margin for the business.
- *The connectivity business does not have gross margins as high as the space business*, but those gross margins are improving, with gross margins jumping from 37% in 2024 to 48% in 2025. This business had positive operating income in 2025, even before capitalizing R&D, and improves substantially with capitalization.
- *The AI business not only has the lowest gross margins of the three businesses, but saw deterioration of those margins in 2025*, reflecting intense competition from other LLMs as well as the rising costs of delivering AI products and services.

There are other parts of the prospectus that come into play in the profitability discussion, with each of the businesses:

- On the space launch business, the cost of launching payloads at SpaceX have been trending down, making its already large cost advantages in the business even larger.
- On the connectivity businesses, there is bad news and good news on the per user front. The bad news is that the revenues, per month, per subscriber, declined from $99 in monthly revenues in 2024 to $66 in monthly revenues in the first quarter of

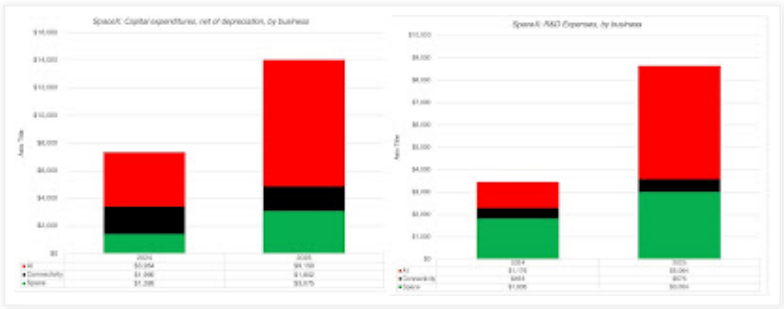
2026. The good news is that the number of subscribers has doubled from 5 million in the first quarter of 2025 to 10.3 million in the first quarter of 2026, with the bonus that the company has been able to improve its profitability (see gross margins in the table above) over time.

- On the AI business, there is not much to go on, on the profitability front, since the focus in the prospectus is more on the increase in compute capacity (see nameplate compute draw on Page 90 of the prospectus) than it is on revenues, especially on the enterprise front. Here again, though, the Colossus lease with Anthropic should help with profitability in the near term.

Story takeaway: The unit economics for the space businesses, in conjunction with the recognition that there are no other substantial operating expenses (outside of the misclassified R&D expense) in either business, lead me to increase my estimate of the target margin for the business to 45%, from 40%. I will leave intact the target margin of 60% for the connectivity business, because once the satellites that service this business are in space, this is the business that will benefit the most from scale. My biggest shift is in my estimated target margin is for the AI business, where the dynamics that are pushing gross margins down, i.e., increased competition and high costs of delivering AI services, will persist; my estimated operating margin drops from 45% to 25%.

3. *Reinvestment*

In my post prior to accessing the prospectus, I did describe SpaceX as a capital intensive business, but the actual spending on capital expenditures and R&D in the prospectus is breathtaking in its magnitude:



In 2025, the company spent almost $14 billion in capital expenditures and almost $9 billion in R&D, a doubling of its reinvestment from 2024. In particular, it is AI that is driving the bulk of this surge, accounting for more than $14 billion in total reinvestment in 2025, with $9.1 billion in capital expenditures and $5.1 billion in R&D. The positive twist that a SpaceX optimist would put on these numbers is that the spending on AI in particular is a positive, indicating that the company is not planning to settle on a niche market strategy, but instead will will go head-to-head

with Anthropic, Google and OpenAI for the enterprise solutions markets. The negative spin is that this ambitious agenda will translate into tens of billions more in capital expenditures in the near years, creating a drag on the cash flows and value destruction if they lose the AI market competition.

Story takeaway: Given that SpaceX is continuing to invest substantial amounts in its space launch and connectivity businesses, I will increase reinvestment in the near term (years 1-5) by lowering how much they will generate as additional revenues for every additional dollar of capital invested (lower sales to capital ratios). With AI, where I was already assuming that reinvestment would be large (with a low sales to capital ratio), the tripling of target revenues will result in a surge in reinvestment to generate the higher sales.

Updating Story and Value

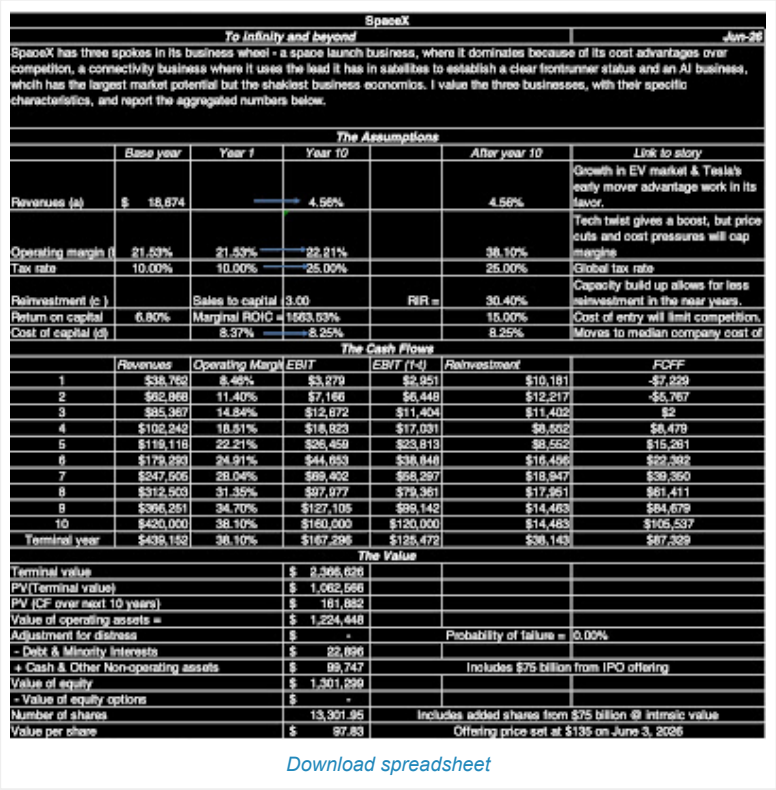
While the core story of SpaceX being a company with growth potential and strong competitive advantages that I framed prior to reading the prospectus remains intact, there are changes to that story that come from the information in the prospectus. The prospectus reinforces the notions that the company is best positioned in the connectivity business to generate both revenue growth and profits in the near term, that its cost advantages in the space launch business will persist and deliver profits, but that target market will be slower to develop, and that the AI business has both the largest target market and poses the biggest challenges, in terms of profitability and capital intensity, for SpaceX.

Bringing together my changes in target revenues, operating margins and reinvestment inputs allows for an update of the input table that I started this section with:

Business	Storyline	Revenues in 2036 ($ millions)	Operating margin in 2036	Sales to Capital (2026-2030)	Sales to Capital (2031-2035)
Space	Dominant player in a growing launch market; 70% market share of $100 billion market	$70,000	45.00%	3.00	4.00
Connectivity	Market leader in satellite broadband, a niche or a very large broadband service market ($1.6 trillion)	$120,000	60.00%	3.00	5.00
AI	LLM competing head-to-head with lead players in a very large, competitive and capital intensice space.	$160,000	25.00%	1.50	2.50
Expansion	Expansion options in space (travel), internet services and AI	$50,000	30.00%	5.00	5.00

Clearly, some of the changes in inputs (such as the higher margins for the space launch business and a bigger target market for AI) will push value higher, and some of the inputs (including a slowing of near term growth for all business, and the much lower margin for the AI business) will push in the opposite direction. Since US treasury rates have risen from 4.20% at the time of my earlier valuation to 4.56% at the start of June, I have increased the costs of capital that I use in the valuation accordingly (to 8.37% from 8.02% to start the valuation, and the steady state cost of

capital to 8.25% from 8.00%; both numbers would put SpaceX at close to the median for all US companies). With these updated inputs, I reestimate the cash flows and the valuation for SpaceX, with the IPO proceeds (estimated at $75 billion) added to the mix:



SpaceX						
To infinity and beyond						*Jun-26*
SpaceX has three spokes in its business wheel - a space launch business, where it dominates because of its cost advantages over competition, a connectivity business where it uses the lead it has in satellites to establish a clear frontrunner status and an AI business, which has the largest market potential but the shakiest business economics. I value the three businesses, with their specific characteristics, and report the aggregated numbers below.						
The Assumptions						
	Base year	*Year 1*	*Year 10*		*After year 10*	*Link to story*
Revenues (a)	$ 18,674		→ 4.56%		4.56%	Growth in EV market & Tesla's early mover advantage work in its favor.
Operating margin (b	21.53%	21.53%	→ 22.21%		38.10%	Tech twist gives a boost, but price cuts and cost pressures will cap margins
Tax rate	10.00%	10.00%	→ 25.00%		25.00%	Global tax rate
Reinvestment (c)		Sales to capital	3.00	RIR =	30.40%	Capacity build up allows for less reinvestment in the near years.
Return on capital	6.80%	Marginal ROIC =	1563.53%		15.00%	Cost of entry will limit competition.
Cost of capital (d)		8.37%	→ 8.25%		8.25%	Moves to median company cost of
The Cash Flows						
	Revenues	*Operating Margi*	*EBIT*	*EBIT (1-t)*	*Reinvestment*	*FCFF*
1	$38,762	8.46%	$3,279	$2,951	$10,181	-$7,229
2	$62,868	11.40%	$7,166	$6,448	$12,217	-$5,767
3	$85,367	14.84%	$12,672	$11,404	$11,402	$2
4	$102,242	18.51%	$18,923	$17,031	$8,552	$8,479
5	$119,116	22.21%	$26,459	$23,813	$8,552	$15,261
6	$179,293	24.91%	$44,653	$38,848	$16,456	$22,392
7	$247,506	28.04%	$69,402	$58,297	$18,947	$39,350
8	$312,503	31.35%	$97,977	$79,361	$17,951	$61,411
9	$366,251	34.70%	$127,105	$99,142	$14,463	$84,679
10	$420,000	38.10%	$160,000	$120,000	$14,463	$105,537
Terminal year	$439,152	38.10%	$167,296	$125,472	$36,143	$87,329
The Value						
Terminal value			$ 2,366,626			
PV(Terminal value)			$ 1,062,566			
PV (CF over next 10 years)			$ 161,882			
Value of operating assets =			$ 1,224,448			
Adjustment for distress			$ -		Probability of failure =	0.00%
- Debt & Minority Interests			$ 22,896			
+ Cash & Other Non-operating assets			$ 99,747		Includes $75 billion from IPO offering	
Value of equity			$ 1,301,299			
- Value of equity options			$ -			
Number of shares			13,301.95		Includes added shares from $75 billion @ intrinsic value	
Value per share			$ 97.83		Offering price set at $135 on June 3, 2026	

Download spreadsheet

The enterprise value for SpaceX edges up from $1.21 trillion, in my pre-prospectus valuation, to $1.22 trillion with the post-prospectus numbers, and the overall equity value increases to $1.3 trillion, with almost all of the increase coming from the influx of $75 billion in cash from the IPO, albeit with a higher share count. The value per share of about $100 will need some revisiting as the IPO numbers firm up and more information is forthcoming on restricted stock units owned by employees, but just as I was finishing this post, a news story hit the wires that the offering price would be set at $135/share.

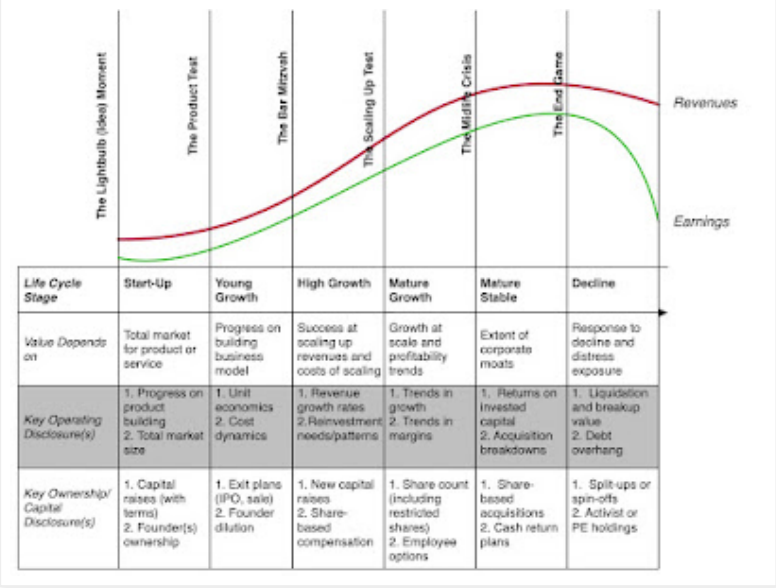
If I were to summarize the impact of the prospectus on my SpaceX story, it would be that it has made the story bigger, but also more volatile. There are a multitude of risks that SpaceX faces in each of its businesses, but the one that I would be concerned about the most is that it will overreach in the AI business, beginning with an overestimate of the target market for AI products and services and the strength of its own competitive position in that market, and following through with investments that reflect those misplaced assessments. Those concerns are heightened by a

voting share structure that locks in Elon Musk's control of the company, since there is little that shareholders can do to restrain the company, if SpaceX doubles down on capital expenditures and acquisitions in the AI space, even after it becomes clear that the AI market is much smaller than anticipated and/or that xAI's offerings are not as good as the competition. If you add to this mix the antipathy that exists between Musk and Sam Altman, you have the potential for a UFC match between two monstrous egos, funded by tens of billions of dollars shareholder money.

Financial Statements and Value: The Life Cycle Effect

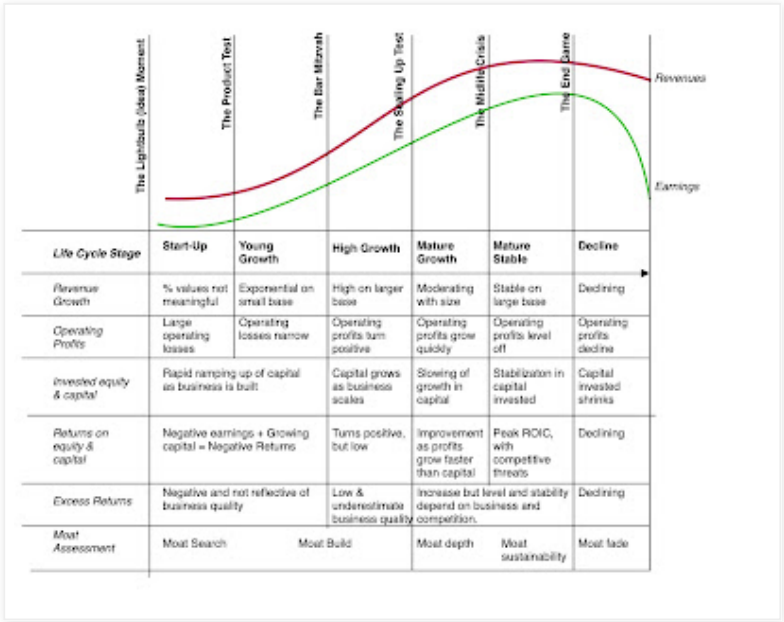
Financial analysis and valuation, going back to Ben Graham's Security Analysis, has always been centered on financial statements, and that focus has become more intense over the last few decades as access to data and analysis tools has expanded. In fact, much of what passes for valuation has become financial modeling, where line items in financial statements are forecast based upon the historical time series, with the proverbial bottom lines being earnings and cash flows. Along the way, ratios computed from financial statement numbers are used to screen companies for investment quality. Some of these ratios, such as accounting returns on capital and equity, have become the basis for assessing company quality and competitive moats in the hands of consultants and investors. The SpaceX prospectus is a case study in why this approach to investing is often myopic and misleading, and why the informational value of financial statements will change as companies grow and mature.

In valuing companies, you are always trying to forecast revenues, profits and cash flows in future, but they key questions you want answered and the drivers of value shift as you move through the life cycle:



Life Cycle Stage	Start-Up	Young Growth	High Growth	Mature Growth	Mature Stable	Decline
Value Depends on	Total market for product or service	Progress on building business model	Success at scaling up revenues and costs of scaling	Growth at scale and profitability trends	Extent of corporate moats	Response to decline and distress exposure
Key Operating Disclosure(s)	1. Progress on product building 2. Total market size	1. Unit economics 2. Cost dynamics	1. Revenue growth rates 2.Reinvestment needs/patterns	1. Trends in growth 2. Trends in margins	1. Returns on invested capital 2. Acquisition breakdowns	1. Liquidation and breakup value 2. Debt overhang
Key Ownership/ Capital Disclosure(s)	1. Capital raises (with terms) 2. Founder(s) ownership	1. Exit plans (IPO, sale) 2. Founder dilution	1. New capital raises 2. Share-based compensation	1. Share count (including restricted shares) 2. Employee options	1. Share-based acquisitions 2. Cash return plans	1. Split-ups or spin-offs 2. Activist or PE holdings

As you can see, for young companies, the key determinants of value include sizing the total market and assessing unit economics, and not the proverbial bottom lines in accounting statements including the magnitude of revenues and profitability. As companies move through the life cycle from start-up to mature to decline, you should expect to see financial statements evolve as well. Young and high growth companies will generally report small revenues (though they expect those revenues to ramp up over time) and losing money and having negative cash flows is a feature, not a bug. As companies mature, revenues will get larger (albeit with lower growth) and profits turn positive, as will free cash flows available to return to shareholders in dividends and buybacks.



Life Cycle Stage	Start-Up	Young Growth	High Growth	Mature Growth	Mature Stable	Decline
Revenue Growth	% values not meaningful	Exponential on small base	High on larger base	Moderating with size	Stable on large base	Declining
Operating Profits	Large operating losses	Operating losses narrow	Operating profits turn positive	Operating profits grow quickly	Operating profits level off	Operating profits decline
Invested equity & capital	Rapid ramping up of capital as business is built		Capital grows as business scales	Slowing of growth in capital	Stabilization in capital invested	Capital invested shrinks
Returns on equity & capital	Negative earnings + Growing capital = Negative Returns		Turns positive, but low	Improvement as profits grow faster than capital	Peak ROIC, with competitive threats	Declining
Excess Returns	Negative and not reflective of business quality		Low & underestimate business quality	Increase but level and stability depend on business and competition.		Declining
Moat Assessment	Moat Search	Moat Build		Moat depth	Moat sustainability	Moat fade

If you allow for the fact that all three of SpaceX's businesses are young, falling in the young to high growth categories, the big questions driving value are about market size and unit economics, since the former provides the basis for revenue growth and the latter determines profitability. That is why, when looking at the prospectus for SpaceX it was the data on total addressable markets, unit economics and capital intensity that had a bigger impact on value, and this information, for the most part, was in the footnotes to the financials, rather than in the financial statements themselves.

For those who are focused on value metrics/constraints (consistently money making, high profit margins and accounting returns) and pricing multiples (low EV to EBITDA or low PE), the SpaceX prospectus is full of red flags. SpaceX is a company with small revenues and large losses, and paying a hundred times revenues for it (which is where a $1.8 trillion pricing would put it) seems foolhardy. I have no quarrels with this point of view, which animates old-time value investing, but this perspective comes with a cost in terms of investment choices. Investors who are wedded to never buying money losing companies or never paying more than twenty times

earnings for a stock will end up with portfolios of mature (and declining) businesses. If that is their comfort zone, the strategy is perfectly defensible, but they should dispense with complaints about never being able to find high growth stocks to invest in or critiques of others who find these stocks attractive, notwithstanding the weak numbers.

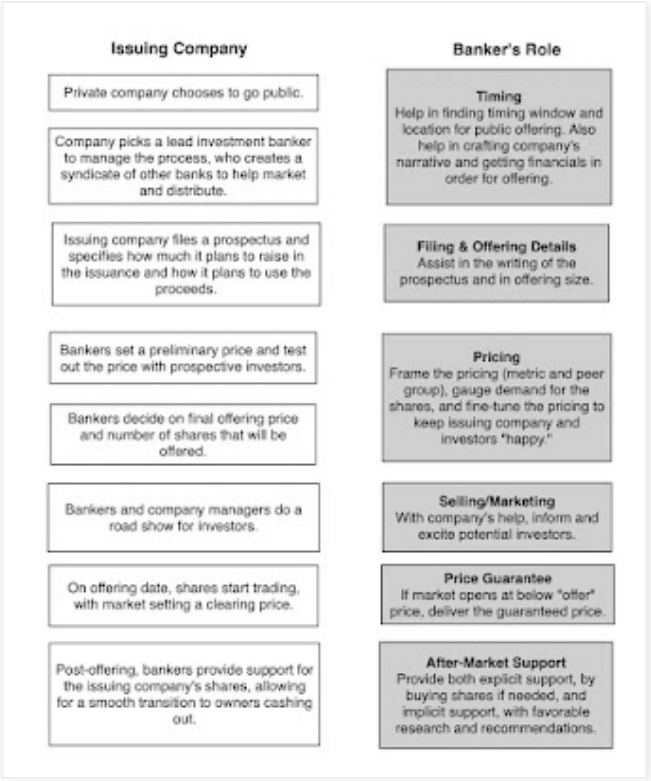
There are many good arguments that can be made about why you should not invest in SpaceX, but basing that conclusion on the fact that they are money-losing or have negative cash flows or trade at a high multiple of revenues is both lazy and unconvincing. In contrast, making a case against investing in SpaceX because you believe that the target markets for its businesses will be far smaller than the company thinks they will be, or that cost and competitive pressures will drive margins down or even that you find its corporate governance structure and dependence on a personality (Elon Musk) off-putting is perfectly reasonable. If you do make that case, though, it is worth remembering that this is your point of view, and that disagreements about market size and profitability across investors, especially in young companies, are natural and healthy. In short, based on my inputs and story, I think that SpaceX is worth about $1.25-$1.3 trillion, but if you contend that it is worth $3 trillion or only half a trillion, it is neither my job nor my place to convince you that I am right and that you are wrong.

The IPO Pricing Game

In the coming weeks, you will undoubtedly be exposed to multiple perspectives on SpaceX, and that is healthy. That said, you will be better equipped to make sense of these perspectives, and perhaps incorporate some of the views into your own, and reject those that do not make sense, if you have an understanding of what an IPO process involves. In particular, understanding the motivations of the different players in the game (the investment bankers setting the offering price and managing the offering, the issuing company, the investors and traders jockeying for shares at that offering price and the traders positioning themselves for the first day of trading) will help determine whether you should be playing this game or sitting it out, at least for the moment.

The Bankers

Let's start with the sequencing that goes into a conventional initial public offering, though alternatives have emerged to it in recent years:



As you look at the role played by bankers to the IPO process, allowing them to keep a slice of the IPO proceeds, the SpaceX IPO is a testimonial to the dwindling value added by bankers on every dimension:

1. Timing: It is urban (or market) legend that investment banks can time markets, and that this market timing can help determine the best time to go public. Just one look at the track record of market strategists at investment banks should dispense with this delusion, since banks (and most institutional investors) are (and have never been) good at gauging market momentum and shifts in mood.
2. Filing and Offering details: It is true that there are technical details and logistical steps to filing a prospectus and setting offering details, but they are almost all mechanical. With SpaceX, I am not sure whether the prospectus, as filed, was the work of a team of bankers, but if it was, I wonder what an entirely Grok-written prospectus would have looked like, and whether we would have noticed the difference.
3. Pricing: In an IPO, the bankers' mission is to price companies for their offering, not value them, and while they usually draw on pricing multiples and peer groups to make that pricing judgment, they are guided by the pricing in the most recent private transactions, usually in the form on venture capital rounds. With SpaceX, that task is simplified by the reality that this company, while private, has had active trading in its private shares, and

that it was priced at roughly $1.2 trillion prior to the IPO process commencing. Adding the $75 billion in offering proceeds, and incorporating the advantages of increased liquidity from being a public company and becoming part of the S&P 500, it is not surprising that there is a sense that the offering will be priced at between $1.5 trillion to $2 trillion, with or without the investment banking input. My guess is that we will end up somewhere in the middle, with some handwaving about revenue multiples and other AI companies used to justify that pricing. (After I finished this post, a news story popped up that the offer price would be set at $135/share, translating into about a $1.8 trillion pricing for the company.)

4. Selling/Marketing: In an age where investment banks have lost credibility and social media is where marketing happens, SpaceX can generate its own marketing spin, and has an army of influencers behind it. In addition, almost every institutional investor has a point of view on whether to own SpaceX or not, it is unclear what exactly a roadyshow can do to augment the sales pitch.
5. Price guarantee: The pricing guarantee that investment bankers offer in initial public offerings is a mostly empty promise, since they systematically set offering prices at below (by 15-20%) what they believe the market will pay. That is the reason that the offer price for SpaceX will be set below the upper end of the range, and while the discount may seem like a significant loss to funders and current owners, the fact that the offering is for less than a tenth of the shares in the company will soften the blow.
6. Post-market support: As a follow-up to the price guarantee, investment banks often offer after-market support for companies in the days after they go public, buying shares if the stock comes under selling pressure. With SpaceX, that option is off the table, since no investment bank has the capital to support the pricing of a two-trillion company, if investors turn negative on it.

In fact, given that banks are perhaps getting more from the initial public offering, in terms of publicity and allotments for their preferred clientele, than SpaceX is getting from their services, you could argue that the bankers should be paying the company for reflected glory, rather than charging them fees. The only good reason that I can think of for SpaceX not going the direct listing route, where you dispense with the kabuki dance of offerings and let the market set the offering price, is that the company needs the cash from the offering, and that route is much more difficult to take in a direct listing.

Issuer (Company, Founder and Investors)

Looking at the IPO from the SpaceX perspective, the public offering will provide benefits. For the investors in the company in its private form, including venture capitalists from early in its life to public investors in more recent years, the IPO will allow them to cash out, albeit after the lock-out period expires in a few months. For the company, the increased access to capital from being a public company will allow it to fund the capital expenditures and investment needs that emanate from the company's ambitions in the AI business. For Elon Musk, the public offering has the

potential to make him the first trillionaire in history, in addition to unlocking new pathways to further enrichment for meeting specified targets (including getting a million people on Mars).

Since some of these benefits have been in existence for many years, the fact that company stayed private for that period is an indication that there are costs to going public that have held it back. The first is that, notwithstanding Musk's voting control of the company, become a public company will open SpaceX to market scrutiny, in the form of earnings reports every quarter and insider trading reports. The second is that the market is fickle, and while it is rewarding companies that invest in AI with high market prices today, it can change its mind and punish them for the same reason. The third is that while there is little that investors can do to trade and make money on overpriced private businesses, they can sell short on public companies.

Investors and Traders

SpaceX is a company that has been in the public eye for a decade or more, even as a privately owned enterprise, partly because of its social media boosters and partly because its space launches make it a magnet for attention. There are many who are drawn to the company, but unable to invest in it as a private business, will now have a chance to do so, if it goes public. But should they try to partake in the initial offering? The answer to that question depends on whether you are an investor, where you buy (sell) companies that you believe are trading under (over) their assessment of value and hope the gap closes or a trader, where you buy (sell) companies where you expect prices to go up (down) in the future, for a multitude of reasons, only some of which may relate to company fundamentals.

I am more investor than trader, and I say that without judgment, since the end game in markets is to make money, not score intellectual points. The truth is that I am not a very good trader, and I am better off staying in my preferred domain, which is valuation, albeit with no guarantees of a payoff. My valuation of SpaceX was driven by my interest in the company and belief that it is in unique, cutting-edge businesses, and my decision on whether to buy into the offering is therefore driven by my assessment of its value. At the rumored pricing of $1.8 trillion for the company, it is too richly priced for my tastes, given my valuation of $1.25-$1.35 trillion for the equity in the company. That does not mean that I will never buy the stock, since the market does change its mind, and if the price does drop by enough, my decision would change accordingly. It is worth remembering that Facebook was selling at half its offering price a few months after its IPO, and that Uber lost more than 50% of its market cap in the year after its public offering, moving both companies from over to under valued.

If you are a trader, though, the game changes. Specifically, the intrinsic value of the company is not central to your decision, perhaps even irrelevant, and your judgment on whether you seek to partake in the SpaceX offering will depend on your reading of market mood and momentum. I would not be surprised in the least to see the offering priced at $1.8 trillion, and see a jump in the price on the day of or in the weeks after the offering, and if that is your most likely scenario, being able to get into the offering at the offer price or even in the first few hours or days of trading will

be a winning strategy. The risk, of course, is that momentum can shift quickly, causing a significant price drop, effectively making timing your trades right key to your trading strategy. The shifting and often unpredictable forces of mood and momentum are also the reason that as an investor, I would not sell short, notwithstanding my value assessment, even if the pricing for the company pushes from $1.8 trillion to $2 trillion or more.

A Loaded Bet on AI!

As the IPO process for SpaceX heats up in the coming weeks, you should prepare yourself for a flood of selling from the company and its bankers, with talk of possibilities and potential dominating the discussion, as well as arguments from the other side, where it will be framed as a vehicle for AI hype, destined to fail. If you are on the receiving end of these sales pitches, you should listen but check the numbers for plausibility and make your own judgments. For the bankers involved and the issuing company, the biggest danger to a successful offering is not that there will be near-term reality checks on their hype, but that the market mood will shift, either in the aggregate or specifically related to AI, in the weeks leading up to the offering. No matter what your views are about the SpaceX IPO, positive or negative, there is no denying that this company is a loaded bet on the AI and Elon Musk, and while that may concern some, there are others who will look at Musk's track record with Tesla and feel the odds are in their favor.

YouTube



Attachments

1. SpaceX prospectus

2. Valuation of SpaceX, post-prospectus on 6/2/26

Blog posts on SpaceX

1. To a Trillion(s) and Beyond: A SpaceX Odyssey!

Posted by Aswath Damodaran at 6:43 PM No comments:
Labels: AI, Big Markets, IPO

Wednesday, May 6, 2026

An Ode to Restraint: Lessons from the Tim Cook Legacy

Through time, we have glorified conquerors and empire builders in politics, civic life and business, from Alexander the Great and Genghis Khan to the tech titans of today. That is no surprise, since these individuals have oversized personas and often change the course of history, but it is also true that this glorification of empire building has shortcomings. The first is the deification of these heroes comes with whitewashing of the dark sides and the costs of empire building. The second is that we discount and undervalue those who make contributions to societal or business advances, but do so quietly and with little fanfare. It is in this context that I was drawn to the story of Tim Cook stepping down as Apple CEO, after a tenure of fifteen years atop a company that has been among the top market cap companies in the world for much of that period. While Steve Jobs, his predecessor as CEO at Apple, has now been deified in business circles, as an unparalleled visionary and business builder, and deservedly so, I think that Tim Cook, in many ways, has played just as significant a role in molding the company into its current day standing, with far less recognition.

Apple's CEOs: From Scott to Jobs to Cook!

Unfair though this may seem, the Tim Cook story at Apple has to start with Steve Jobs. Jobs co-founded the company in 1976, with Steve Wozniak, and while the company went through a series of CEOs in the next two decades, Jobs was the face of the company in its early years. While it is easy, with the benefit of hindsight, to view these as good years for the company, those early years reflected both Job's strengths and weaknesses. His vision and force of personality gave rise to the personal computer in its current form, as a tool for everyone to use, not just tech geeks, and as someone who bought his first Mac (the 128K without a hard drive) in 1984, and has stayed a Mac user since, I am grateful. That said, the dark side of Jobs, manifested in impatience with underlings and an obstinate belief that he knew what customers needed better than they did, led to the Lisa, the only Mac I regretted buying almost immediately after my purchase, and a loss of business markets to Microsoft. Those failures led Apple to the brink of failure, and to Jobs being cast out of the company by its board in 1985, though the CEOs that

followed had neither the strategic vision nor the business-building capacity to rescue the company.

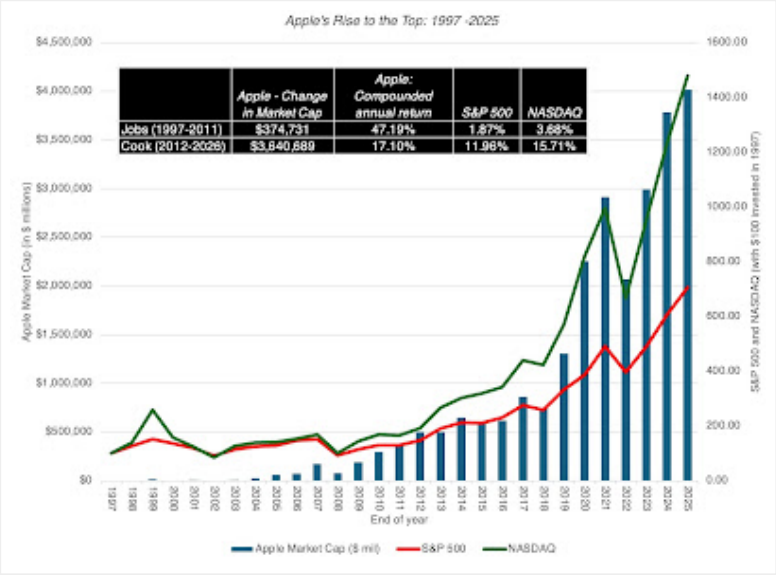
In 1997, Apple looked like it was a company heading into oblivion, as Windows became the dominant operating system for personal computers, and it seemed like Apple had lost its purpose. The August 1997 return of Steve Jobs,, who had used his years in the wilderness to build Pixar, a company that revolutionized animated movie making, is now the stuff of legend, as he rebuilt Apple in the ensuing years into a powerhouse, around the iPod, the iPad and most of all the iPhone. While there are books and movies chronicling the Steve Jobs success story, it is worth asking what the difference was between the first iteration of Steve Jobs at Apple (from founding to leaving in 1985), where Apple lost ground to Microsoft, and the second iteration of Steve Jobs (from his return in late 1997 until his resignation in 2011). The first was that *he was older,* and to the extent that with age comes some wisdom, it helped, but it is unlikely to have been the change maker. The second was that in his period away from Apple, J*obs created and built up other companies, with Pixar being the biggest*, where he learned to deal with people better and perhaps compromise a bit more than he used to. The third was that he benefited from *the presence of Tim Cook*, first as an executive in Apple sales and operations, and more importantly, as chief operating officer (COO) for Apple, starting in 2005. If Steve's skill was vision, where he showcased Apple's next "big innovation" at meetings in his trademark black turtleneck, Cook's skill was building manufacturing hubs and supply chains to convert the vision to products. That separation of vision from business building created the Apple juggernaut in the first decade of this century. While that division of labor clearly was in the company's best interests, Jobs deserves credit for being willing to set his ego aside and delegate the powers to make it happen.

Tim Cook has been CEO for fifteen years, and when he retires on September 1, 2026, *he will have been the longest serving CEO at Apple*. It cannot have been easy, especially in the early years, as the comparisons to Steve Jobs were front and center, and there was pressure on him to continue in the same path. To Cook's credit, he never tried to be Jobs, and he created a very different template for himself, one that fit him and the company well, and served as a testimonial to his self assurance. In one of my talks about a decade ago about Apple, I described Cook, perhaps harshly, as a man without a visionary bone in his body, but one who would make sure that the trains ran on time (or the iPhones were delivered as promised), and I think that he has used that strength to good effect during his years as CEO of the company.

Apple's Finances in the Twenty First Century: The Steve Jobs and Tim Cook Years!

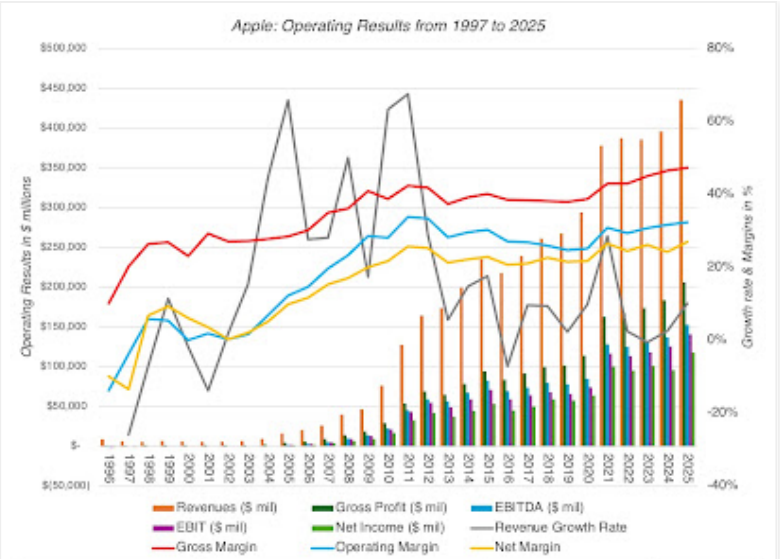
Steve Jobs handed over a company to Tim Cook in 2011, that was extraordinarily profitable, and at the time of his leaving, already the largest market cap company in the world. While that fact leads some to discount what Cook has done at Apple since, I think it is worth going back in history and looking at corporate handoffs of great companies, and how often they become tangled messes, as new CEOs overreach and overpromise.

The place to begin our comparison of the Jobs and Cook tenures is by charting Apple's market capitalization, with the delineation into the Jobs years (1998-2011) and the Cook years (2012-2026):



Looking across the aggregated years across both CEOs, it has been an extraordinary time. Apple began the Jobs tenure as CEO with a market cap of $1.68 billion, and by the end of 2025, its market cap had risen to over $4 trillion, and its performance burnishes the reputations of both Jobs and Cook. Jobs provided the foundational boost for the company and the innovations he presided over delivered a *compounded annual price appreciation of 47.19% between 1997 and 2011,* a period when US equities were struggling and Apple reached the top of the market cap table in 2011. With Tim Cook at the helm, Apple added an astounding $3.64 trillion in market cap, but a strong equity market provided strong tailwinds, and the company's annual returns were more modest. On a percentage return basis, the Jobs years were better, but in my view, the fact that the annual returns in the Cook years were just as impressive, because they had to be earned on a much larger firm.

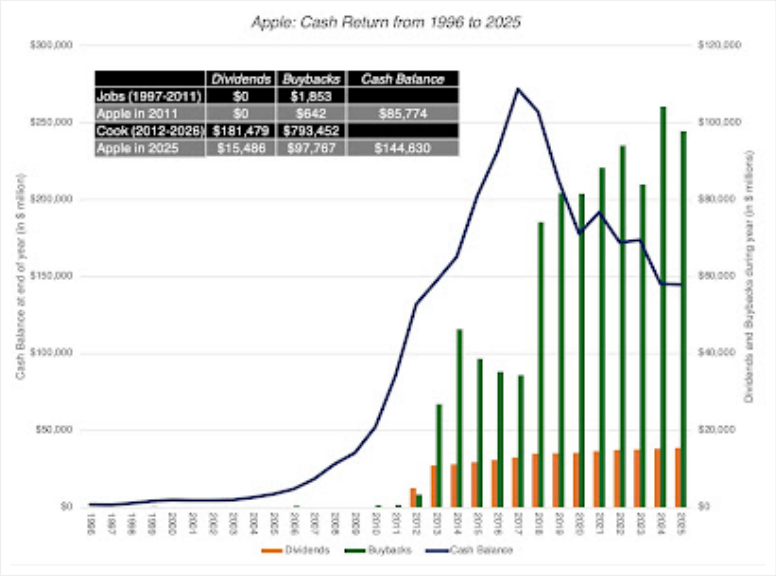
The reasons for Apple's sustained increase in market capitalization were simple - *solid revenue growth and a profit machine that delivered high margins,* even as the company scaled up:



	Revenue: CAGR	Gross Margin	Operating Margin	Net Margin
Jobs (1997-2011)	23.67%	36.56%	23.48%	17.86%
Apple in 2011	28.82%	41.91%	33.46%	25.35%
Cook (2012-2026)	8.52%	41.91%	29.30%	23.88%
Apple in 2025	10.07%	47.33%	32.38%	27.04%

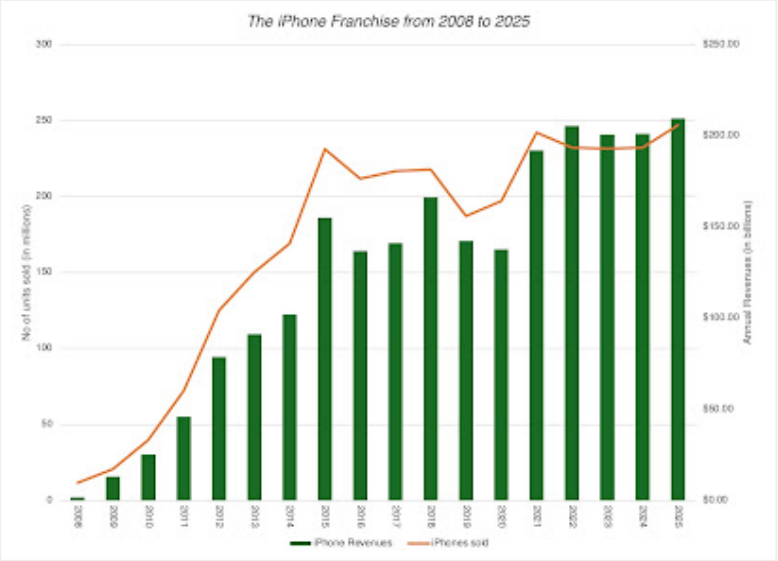
As with the market capitalization comparisons, this chart yields metrics that are favorable to both Jobs and Cook. Under Jobs, the company scaled up its revenues significantly, with a compounded annual revenue growth rate of 23.67% between 1997 and 2011, and just as significantly, went from posting subpar margins and a net loss in 1997 to becoming one of the most profitable tech companies in the world, Under Cook, revenue growth rates came down (to a compounded annual average of 8.52% between 2012 and 2025), but on a much larger scale, and the company preserved and grew its profit margins.

There was one corporate finance dimension on which Cook deviated from Jobs, and that was on cash return or dividend policy. In the chart below, I look at the cash returned to shareholders by Apple during the tenures of the two CEOs:

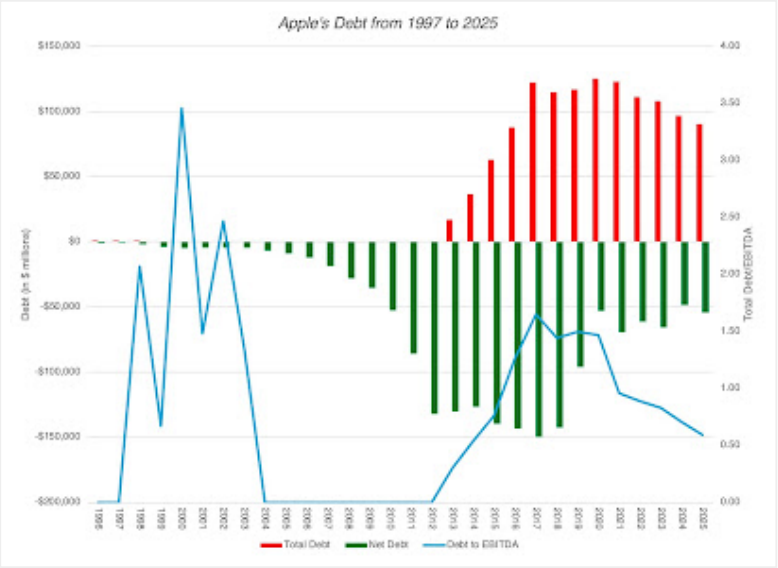


During Job's tenure at Apple, the company paid no dividends and initiated only modest cash buybacks, mostly to cover stock-based compensations. With Tim Cook as CEO, Apple was one of the greatest corporate cash success stories of all time, initiating dividends in 2012 and increasing them over time, and supplementing those dividends with cash buybacks that, in the aggregate, were the largest in corporate history. *In sum, the company has bought back almost $800 billion between 2012 and 2025, and the most astonishing feature was that, while returning all of this cash, the company also accumulated one of the largest corporate cash balances in history.*

To the question of how Apple was able to return this much cash, increase its cash balance and still grow itself, the answers are three fold. The first is that the iPhone, perhaps the most valuable single product in business history, continued to deliver for the company, with modest reinvestment needed on its upgrades.

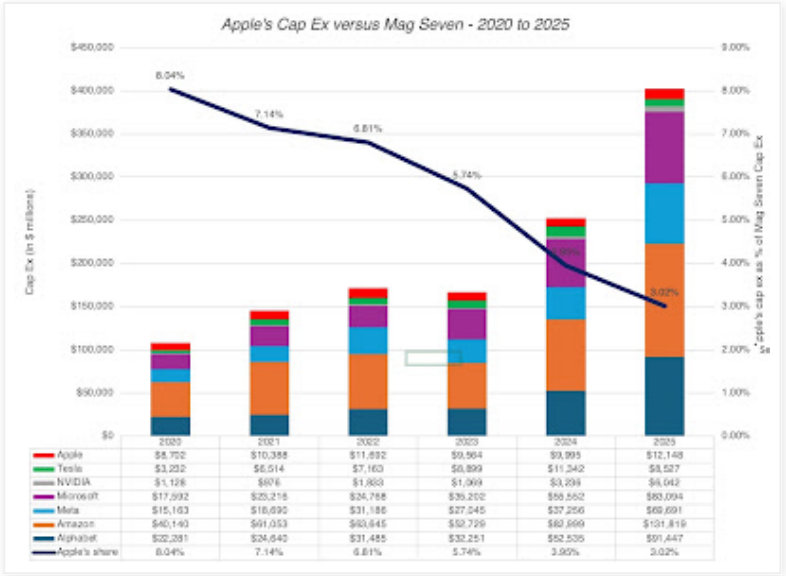


While much of the credit for the iPhone is still given to Steve Jobs, and rightly so for fostering the innovation, credit is also due to Cook, who has taken the franchise handed to him, and grown it on steroids. The second is that the company borrowed $17 billion in 2013, a Cook departure from a Jobs practice of avoiding debt, and it has added to that debt load over time, though it remains a small slice of overall value:



While much is made of Apple's debt foray, it is worth recognizing that *Apple is still a very lightly indebted company on any debt metric,* and that if you net the company's considerable cash balance out against its total debt, *its net debt has always been negative (cash exceeds debt)*. In fact, Apple's use of debt is so light that the only rationale for its existence is creating a presence in the bond market, just in case it needs to use it more in the future. The third feature is that the company has been cautious in its forays into new products and markets, especially outside its domain, and this shows up in two data series.

- The first is that while Apple has acquired more than a hundred companies, *almost all of them are small, private technology companies with small price tags*, with the intent being bringing their products and services into the Apple ecosystem after the acquisition. In fact, its largest acquisitions during this century are so small that they represent petty cash, relative to its cash balance as a company. Beats, for instance, which was one of Apple's biggest acquisitions cost the company about $3 billion, a number dwarfed by its cash balance that year, which was more than $100 billion.
- The second is that in the last five years, as big tech companies have gone on an AI capital expenditure binge, *Apple has been the outlier, holding back on its AI investments*, and this can be seen in the chart below, where I compare Apple's capital expenditures to those of the rest of the Mag Seven:



  As the rest of the group has ramped up its capital investments, with much of it going into AI, Apple has held back, and its share of the total cap ex at the companies has fallen from 8.04% to 3.02% over the period.

In sum, looking at the changes at Apple over the last fifteen years, t*he company has changed from the growth engine, driven by disruptions, in the Jobs years to a mature, cash-returning and*

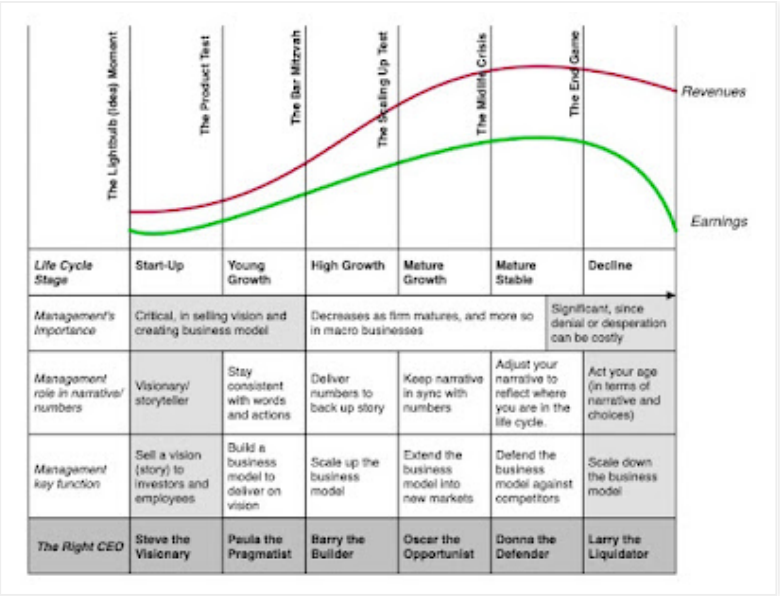
more cautious company under Cook. I have posted more about Apple than about any other company in the world (and I have a sampling of some of those posts at the end of this post) and have been a shareholder in the company for significant portions of both the Jobs and Cook tenures. I have not always agreed with either man, on choices that they have made at the company, but I respected both of them enough to view them as good stewards of my investment. In a world full of CEOs who are quick to herd to what the consensus view is, I admire both men for their willingness to stand on their beliefs.

Vision or Restraint: A Life Cycle Perspective

If you were to create a profile of Tim Cook, the manager, based upon the choices that he has made at Apple during his tenure as CEO, two very divergent views emerge. To his admirers, his actions on some fronts (initiating dividends, massive stock buybacks, borrowing money) and inaction on other fronts (no big acquisitions, diffidence on AI investments), *represent an exercise in discipline and restraint,* preserving the company's crown jewel (the iPhone) and fending off the bankers and consultants, with their false promises. To his critics, and there are quite a few, Cook's caution has cost Apple its disruptor status, when it could have used its ample cash reserves to buy its way or invest in into almost every new business that has bloomed in the last fifteen years. In fact, they point to chances that Apple has had to buy some of the biggest stars in the market, from Tesla and Netflix more than a decade ago to Anthropic, Mistral and Perplexity in more recent years.

It is impossible to argue that one side is right and the other side wrong, but it is undeniable that both pathways (the restrained pathway that Apple adopted and the more aggressive pathway that it could have taken) include trade offs. It is true that Apple's restraint has led it to miss out on some of the biggest trends in technology over the last decade, but it has also avoided the overpayment that is so common with high profile acquisitions of big companies. The argument that Apple would be worth a lot more today if it had bought Netflix or Tesla a decade ago falls flat for two reasons. The first is the selection bias in picking two companies that, in hindsight, have emerged as winners, when in fact there were at least a dozen other worse-performing companies that were also on Apple's radar. The second is the presumption that companies like Tesla or Netflix would have been just as successful, owned by Apple, as they were as stand alone enterprises. The clash of corporate cultures that would have ensued if Apple had bought either Tesla, a company that reinvents its business narrative every few hours, or Netflix, an entity that makes content in quantity with the hope that some it sticks, would have been epic, with the risk that both Apple and its acquired target would have gone down in flames.

More generally, though, the question of whether you want a visionary or a disciplined business builder at the top of a firm is not one that has an easy answer, since it depends on the firm in question. In my work on corporate life cycles, I focus on the management skills that are needed most in a company, based upon where it is the life cycle, and that may help address the choice between vision and restraint:



Life Cycle Stage	Start-Up	Young Growth	High Growth	Mature Growth	Mature Stable	Decline
Management's importance	Critical, in selling vision and creating business model		Decreases as firm matures, and more so in macro businesses			Significant, since denial or desperation can be costly
Management role in narrative/ numbers	Visionary/ storyteller	Stay consistent with words and actions	Deliver numbers to back up story	Keep narrative in sync with numbers	Adjust your narrative to reflect where you are in the life cycle.	Act your age (in terms of narrative and choices)
Management key function	Sell a vision (story) to investors and employees	Build a business model to deliver on vision	Scale up the business model	Extend the business model into new markets	Defend the business model against competitors	Scale down the business model
The Right CEO	Steve the Visionary	Paula the Pragmatist	Barry the Builder	Oscar the Opportunist	Donna the Defender	Larry the Liquidator

With young companies, **vision dominates,** as managers work to sway investors, employees and nascent customers that their product or service will find a market. As the vision takes hold, converting it into commercial products and services requires trading off some portions of vision for **pragmatism,** *in the interest of getting the business going***.** As products and services find demand among customers, **business building** becomes a key difference-maker, with the grunt work of marketing, production facilities and supply chains coming into play. Assuming that you have made it through these three stages, the trade offs of scaling up come into focus, and as you hit market limits, success depends on **being opportunistic** in finding new products and markets, but only if they exist. In corporate middle age, pathways to easy growth, especially at scale, become difficult to find, and to the extent that value comes from moats and core products, **playing defense** against competitors takes priority. Finally, in decline, a phase that no company ever wants to enter, but is inevitable at some point, you need to be willing to **shrink a firm**, shutting down businesses that no longer deliver value and selling other assets to high bidders.

Given these very divergent management functions, it should come as no surprise that there is no prototype for the perfect CEO, McKinsey and Harvard Business School blueprints notwithstanding. Viewed in this framework, I would argue that Apple has been lucky with its last two CEOs, both in terms of persona and in terms of sequence. When Steve Jobs rejoined Apple in 1997, the company had hit rock bottom, and with little to offer in liquidation, his vision allowed for a reincarnation, with disruptions leading the way, and as we noted earlier in this post, having a strong chief operating officer in Tim Cook made the difference. The Apple that Tim Cook inherited, when he became CEO, was a very different entity, already the world's largest market cap company, with a superlative franchise in the iPhone. In corporate life cycle terms, Apple was a mature growth company, and what Cook lacked in opportunism , he made up for by defending Apple's biggest product line(iPhone) and augmenting value with increments like the app store and

devices. That said, while each of these men created value for shareholders, I don't think that either would be regarded as highly, if you swapped their tenures in terms of timing. I don't think Tim Cook would have been able to bring Apple from its near-demise to being on top of the corporate universe, if he had become CEO in 1997, and I think Steve Jobs would have been ill-suited to the Apple that was in existence in 2011.

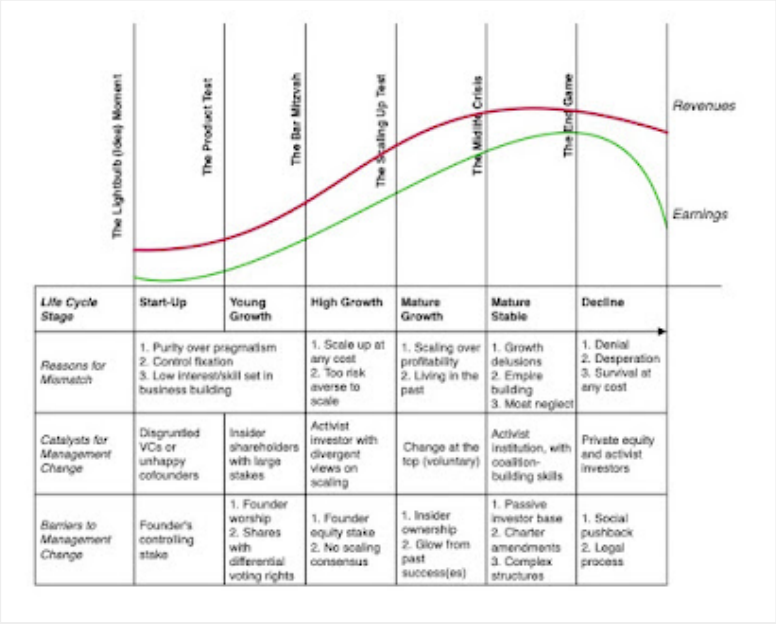
Aging, Management Mismatches and Corporate Governance

In a post from a few years ago, I used the connection between CEO type and corporate lifecycle to examine why management mismatches occur at firms, and the consequences of that mismatch. Specifically, there are three confounding factors that can make matching up CEO to company, given where it is in the life cycle, complicated:

1. Like humans, companies age, but unlike humans, **the rates at which different companies go through the life cycle can be wildly different**. An infrastructure or manufacturing company can take decades to become operational, followed by extended phases of growth and maturity, before going into decline. In contrast, a tech company can have explosive growth early in its life, spend a brief period enjoying the fruits of its success as a mature company before declining precipitously. As a consequence, managers and investors who use chronological age as their corporate aging metric can misjudge where they are on the life cycle.
2. While aging is inevitable for both humans and businesses, some mature or even declining **businesses can find pathways, either through happenstance or management choices, to rediscover their youth**. These businesses become the stuff of legend, and they are the subjects of books and business school case studies, and their CEOs are elevated to management deities.
3. The narratives built around companies that reincarnate and the CEOs atop these companies also **feed into management incentives and behavior.** The story of Steve Jobs at Apple has been told and retold, but it is worth remembering that for every story of reincarnation, there are a hundred stories you can tell about other CEOs who tried to follow the Apple playbook, spending billions on reinventing their companies, with little to show in terms of payoffs. (See my posts on Marissa Mayer at Yahoo! and on Blackberry.) In essence, the glorification of CEOs who bet big on turnarounds at mature or declining companies, and win, sets up CEOs facing similar circumstances to behave like riverboat gamblers, when making management choices at their firms. After all, if their bets pay off, they join the legend crowd, and if they do not, they contend that they did their best, and that circumstances conspired to bring them down.

The bottom line is that there are a number of ways in which you can end up with CEO mismatches - a CEO who cannot adapt to the changing demands of an aging business, a hiring mistake or even changes in the macro environment, and when those mismatches occur, is is inevitable that there will be friction between the CEO and shareholders. In a sense, almost all

corporate governance challenges can be traced back to management mismatches, and the power (or the absence of it) that shareholders have to fix those mismatches:



While Tim Cook's time as CEO of Apple is now seen through rose-colored lens, it is worth remembering that Apple was targeted repeated early in his tenure by activist investors. While some of the changes that these activists were pushing for were warranted, some were not, and Cook deserves credit for not capitulating. Carl Icahn, for instance, wanted Apple to increase its debt substantially, borrowing hundreds of billions, but I took issue with his argument that Apple could borrow this money at the low rates that he was extrapolating. A couple of years later, David Einhorn made his play, arguing that Apple should issue preferred shares with a 4% dividend yield, and I noted that preferred stock could bring with it all of the cashflow commitments of debt, with none of the tax advantages. I have long argued that the best defense a management has against activist investors is delivering superior performance and returns, and Tim Cook delivered on both dimensions, and faced little more than sniping from disgruntled investors in his later years as CEO. In the last four years, the criticism has come primarily from analysts who fault his caution, and argue that Apple risks falling behind its more aggressive competitors in the AI race, but here again, Cook has stood his ground.

Management Transitions, Past and Present - The Mag Seven

I don't envy John Ternus, who is Cook's heir apparent, because he is following two CEOs who were immensely successful, albeit in different ways. If there are lessons he can learn from both Jobs and Cook, they include the following:

- Find your own path: There will be pressure from some investors to be just like Jobs, and go for big disruptions, or from others to imitate Tim Cook, and leave Apple as a

cash machine. While it may take time, Ternus has to find his own path as CEO, based on not only what he brings to the table, given his background in computer hardware, but on what Apple's strengths are as a company in 2026 and the markets it is facing right now.

- Adapt to the company you are managing: Just as the Apple that Jobs took over in 1997 was very different from the Apple that he handed over to Cook in 2011, the company that Ternus takes over is different from the ones handed over in either of the prior iterations. When you are at the helm of one of the largest market cap companies in the world, you have to start with the recognition that any new product or service that you introduce will have to be huge to make a dent in the operating metrics (revenues and profits) or market capitalization. In addition, the franchise that holds up the company's cash machine is the iPhone, and Ternus cannot afford to take his eyes of that prize.
- Keep the feedback loop open: When you are a company worth trillions, with legions of shareholders, and hundreds of analysts, you will have advice meted to you constantly on what you should or should not do. Much of that advice will be bad, and should be dismissed, but some of it is worth listening to and perhaps converted into policy. In addition, as CEO, I hope that Ternus views the market price as a crowd judgment on Apple's actions rather than the product of speculation, and accepts that while that judgment can be wrong, it should be taken seriously.

I wish Mr. Ternus the best, for purely selfish reasons. As a Mac and Apple device user, I want the company to prosper and continue to make products that I can continue to use on a daily basis, and as a shareholder, I want my investment to do well.

The attention, in this post has been on the management transition at Apple, but management transitions are part and parcel of every company, with the changes sometimes forced on the company and sometimes voluntary. Expanding the discussion of management to the other companies in the Mag Seven can provide us with an opportunity to examine management transitions that have either already happened or that will happen in the future, and the ensuing frictions:

- **At Microsoft**, the only company in this group that traces its vintage back to Apple, there have been two CEO transitions, from Bill Gates to Steve Ballmer in 2000, and from Ballmer to Satya Nadella in 2014. While Gates built Office and Windows into cash cows, and Ballmer preserved them, Nadella created his own pathway to reincarnation by building up a cloud business that is now the dominant source of revenues for the company. By partnering early with OpenAI on LLMs, and investing massively in data centers, Nadella is now making a bet that AI can provide a further boost to the company's operations, perhaps setting the stage for a second rebirth.
- **Amazon has seen a management transition**, where a legendary founder (Bezos) left the firm in 2021, and his successor (Andy Jassy) has taken the reins, with remarkably little fanfare. Like Nadella, though, Jassy is betting big on AI being a

growth and value driver, and the success or failure of that bet will largely determine how his stint as CEO gets judged.

- **Alphabet offers a case study of a company that tried to split the difference**, by separating its cash cow (Google advertising) from its other businesses, naming Sundar Pichai as the CEO for Google, while remaining atop the other Google businesses (the bets in Alphabet). That experiment has struggled to deliver, as the other businesses remained earth-bound and in 2019, and Pichai took over as CEO of Alphabet as well. Over its lifetime, Alphabet has been immensely successful in coming up with products and services that catch public attention, whether it be its development of the Android operating system or its work on Waymo or Gemini, but it has struggled to convert those successes into revenues and operating profits.
- I**n three of the companies (Tesla, Meta and Nvidia), founders remain CEOs, though they bring very different perspectives and personalities into their roles**. As I noted in my last post on SpaceX, Musk has veered between genius and eccentricity in his stewardship, but shareholders at Tesla have largely benefited from the rollercoaster ride. At Meta, Zuckerberg has been a shrewd businessperson in his management of his social media holdings, with savvy acquisitions of Instagram and Whatsapp boosting his ad-driven ecosystem, but he has also been headstrong in his pursuit of ventures that he feels are the "next big thing". His expensive failed bet on the Metaverse led some investors to question his governance, and many of these investors worry that his bet on AI will play out similarly. Finally, on Nvidia, the company's soaring market capitalization and huge success with AI chips has pushed Jensen Huang into the spotlight, but less than a decade ago, there were questions about his management as well.

The fact that **all six of these companies have invested heavily in AI is a lead-in to what could be the key test for management at all of them**. If the AI investments pay off and deliver value, Nadella will cement his legend status, Jassy will have created his own legacy at Amazon, the Alphabet experiment will finally pay off, and the founder-run companies will have more room to run. If the AI investments fail, though, Nadella's reincarnation reputation will take a hit and Jassy's position atop Amazon will be at-risk. The AI failure will also raise doubts about Alphabet's capacity to grow beyond advertising and the rumblings about Zuckerberg's big bets will get louder, but at these two companies, it is unclear what investors, no matter how large their holdings are, can do, since they have acquiesced to a voting share structure at these two companies that has reduced them to bystander status. At Alphabet, Brin and Page control 51% of the voting rights, with less than 10% ownership, and at Meta, Zuckerberg controls 57% of the voting rights, with about 13% of share ownership.

An Ode to Restraint

While there are many who compare to Tim Cook to Steve Jobs and find him wanting on vision and flair, I am grateful, as an investor in Apple, for the restraint and discipline that he brought to the job. That gratitude will stay intact even if Apple's caution on AI turns out to be a mistake, since

the restraint and rectitude that Cook brought to his job are management qualities that significantly undervalued. I don't teach from or write cases, but I would love to see more business school cases about CEOs like Cook who are not easily swayed by the temptation of more growth and ego-driven acquisitions. I loved the Steve Jobs movie, but I don't expect to see a Tim Cook movie anytime soon, and while that is understandable, it also explains why we will continue to have too many CEOs at companies viewing themselves as saviors, gambling shareholder money on turnarounds and rescues, when the better pathway would be acceptance and shrinkage. I believe that investors lose more money from companies trying to do too much rather than from them doing too little, and from overreaching than from underachieving.

YouTube Video



An Ode to Restraint: Lessons from the Tim Cook L

Aswath Damodaran





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10. Love the company! Love the product! Love the stock! (September 9, 2013)
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My book on the corporate life cycle

1. The Corporate Life Cycle

Posted by Aswath Damodaran at 5:47 PM No comments:
Labels: CEO, Corporate Life Cycle, Management Transition

Thursday, April 23, 2026

To a Trillion(s) Dollars and beyond: A SpaceX IPO Odyssey!

In *2001, A Space Odyssey*, a movie that was well ahead of its time when it was released in 1968, Hal (the computer) famously responded to questions about his reliability with "it (mstakes) can only be attributable to human error". I was reminded of my fallibility repeatedly as I tried to value SpaceX ahead of its initial public offering, a market debut that is shaping up as a barn-burner for three reasons. The first is that in a market where there are many young companies all trying to claim to be futuristic in their offerings, SpaceX clearly stands out as the real thing, with rockets, satellites and AI all residing under its corporate umbrella. The second is that its founder (Elon Musk) is the richest person in the world, has upended one legacy business (autos), bought a social media company as a soapbox and made his presence felt on the the political stage. Love him or hate him, Musk is definitely not boring, and his capacity to spin business narratives that seem outlandish at first hearing. but become conventional wisdom later, clearly adds to the allure of SpaceX. Third, if the private market pricing feeds into the public offering, SpaceX could very well become the most valuable IPO of all time, joining the rarefied list of trillion-dollar companies, on listing. That said, I may be getting a little ahead of the game here, because SpaceX has not filed a public prospectus yet, and little is known about its financials other that drabs of information that have been leaked to the press. I will forge on, nevertheless, with the stipulation that this is a first iteration, and that I will revisit it, as more information comes out about the firm's financial standing and its IPO plans.

The History of SpaceX

You may be surprised to hear that SpaceX is older than Tesla, at least in terms of chronological age, founded on March 14, 2002, in El Segundo, California. At its founding, Musk stated its goals as reducing the costs of space transportation and travel to Mars, but was viewed as having little chance of success by the space establishment, composed then of government agencies (NASA) and a few defense firms (Boeing and Northrop Grumman). SpaceX applied the lessons of modular engineering from the software business and it launched Falcon 1, its first space launch vehicle in September 2008; that successful launch led to a NASA contract for $1.6 billion, and rescued the company from near bankruptcy. In subsequent years, SpaceX developed Falcon 9, a reusable and heavier vehicle, with the Dragon Spacecraft unit, and became the first commercial entity to deliver cargo to the International Space Station. In 2013, SpaceX launched its first mission for a private customer, and quickly secured a dominant market share of commercial launch contract market. In recent years, SpaceX has invested in an even more ambitious version (in terms of size and power) of reusable spacecraft with Starship, and while its first launch in 2023 exploded in space, the company is clearly moving towards making it functional.

Along the way, the company added to its business mix, first with The Boring Company, a company that specialized in building tunnels that could be used to transport people, in 2017, before spinning it off as a separate entity. More significantly, in 2019, the company launched sixty Starlink satellites, with the end game of offering satellite-based internet services to customers, especially in areas where conventional internet service was limited. That endeavor has now grown to include thousands of satellites and had more than ten million active subscribers spread across the world, at the end of 2025. In February 2026, the company created its third business arm, with its acquisition of xAI, the parent to Grok, the Musk-developed competitor in the LLM space.

SpaceX had a slow start financially, as its initial years were spent developing the Falcon 1 rocket, and even after that development, the dependence on the US government and commercial satellite launchers resulted in revenues growing much more slowly than they did at Tesla, Musk's other high-profile creation. Even as late as 2021, SpaceX reported revenues of just over $2 billion, almost entirely from its launch business, but Starlink's subscriber based model has allowed revenues to increase more than five-fold since, reaching an estimated $15.6 billion in 2025, with just under 30% coming from the launch business ($4.1 billion) about the rest from Starlink subscriptions and related businesses ($11.4 billion); xAI, which was acquired in 2026, had subscription revenues of roughly $100 million in 2025. Without full financials to back up the statement, it is estimated that SpaceX generated an EBITDA of $8 billion in 2025, though with depreciation and other expenses considered, it is not clear how much (if any) operating (or net) profits the company delivered during the year.

On the funding front, Elon Musk used a portion of his winnings ($180 million) from his PayPal exit as seed money ($100 million) for founding SpaceX, but the company has required multiple rounds of venture capital to fund its infrastructure needs. The first venture capital round of about $12 million was in 2002, but there have at least thirty additional infusions amounting to more than $12 billion. While SpaceX counts big name venture capitalists in its investing roster (Founders

Fund, Andreessen Horowitz and Sequoia), it has also seen increasing investments from public equity investors such as Fidelity and public tech companies such as Google. *While these venture capital investments have diluted Musk's ownership over the years, he continues to own about 42% of the equity in the company, and with differential voting rights, close to 80% of the total voting rights in the company*.

Valuing SpaceX

It is true that intrinsic valuation, at least in its discounted cash flow avatar, is much easier to do at companies that have many years of historical data and peer groups of companies in the same business, and there are some who view one or both as pre-requisites. If you adopt that point of view, it is easy to see why so many view SpaceX as a company that cannot be valued (yet), since you don't have access to even a single year of financials, let alone a long history, and there are no true competitors. In fact, it is likely that even if the financial statements are made public in a prospectus, most will continue to avoid valuing the company, using uncertainty about the future as an excuse. *If you define intrinsic value as the value of a business based upon its capacity to generate cash flows in the future, there is nothing in that definition that requires either historical data or peer group information, and statistically, the fact that you face uncertainty or that you are missing information does not imply that you cannot make estimates, just that the estimates will be noisier.*. I have long argued that you can estimate the value of young companies with minimal data, as long as you build a valuation around a business narrative, and accept that this valuation will change, as circumstances do, and with SpaceX, I will get a chance to put this argument into practice.

To value SpaceX, I consider each of its three core businesses separately since they differ not just on operating metrics, but also on the competition faced in each one.

- The **launch business**, which is where SpaceX was born, is still its most identifiable business, and frames the company's story not just as a futuristic company, but one that was able to overcome some of the most significant technological challenges of any start up and not just survive but thrive. SpaceX has established such a robust and long-standing cost advantage over its competitors in the business, stemming from its existing infrastructure investments and reusable rocket technology, that it had a market share in excess of 80% of the launch market in 2025. Its competition will come from some private and government-funded players, who may be able to capture market share, notwithstanding their higher costs, due to security and nationalistic concerns..
    - *Valuation narrative: The space launch market is estimated to be about $30 billion in 2026 and is expected to grow to $100 billion in 2036, as government and private business demand increases. SpaceX will continue to dominate the business, albeit with a slightly less dominant market share (70%, down from >80% in 2025) of the total market and as costs decrease with scale, operating margins will increase over time to 40%. (I am being*

conservative in my estimates, insofar as I am ignoring space travel and expanded business opportunities in space, but I don't think, at the moment, that either offers a viable path to augmenting revenues).

- The **internet service business**, built around Starlink, is the business that accounts for almost two thirds of the revenues of the company in 2025, and it builds on the infrastructure built for the launch business, since SpaceX has used it to launch thousands of satellites into space. At the end of 2025, Starlink had close to 10,000 satellites in space, about two thirds of the entire global count, and is adding to that number every month. That has allowed it to double its subscriber numbers to just over 10 million, in the last year, and while Amazon's acquisition of GlobalStar has brought a potential competitor into the mix, GlobalStar has a fraction of the satellites that Starlink does. The challenge for any satellite-based internet service provider is that notwithstanding the use of low-earth orbit (LEO) satellites to improve service, the broadband service lags more conventional internet technology (fiber optic and cable) in much of the world, leaving it (at least for the moment) with a niche market of rural areas, countries with damaged or no infrastructure and people on the go (airplanes, trains and cars). Thus, while the total internet service market is estimated to be close to a trillion and a half dollars globally, in 2025, satellite-based service accounted for about 1% of that market, delivering under $15 billion in revenues, with StarLink having a dominant market share.
    - *Valuation narrative: The satellite internet services market will continue to grow, as technology improves service quality and transit demand for better wifi grows, from $15 billion to $160 billion (from less than 1% to 10% of of the internet service market) over the next decade. Starlink will see more competition, but its lead in satellites and capacity to use its launch business to get more into space, will give it a substantial advantage and a market share of 75% of the overall market). The cost of customer acquisition will ease over time, as business customers become a larger portion of the business, and operating margins will approach 60% in steady state, as the unit economics are very positive.*
- The L**arge Language Model (LLM) business**, from the acquisition of xAI, has brought AI into the SpaceX story, and while that may add to the pricing excitement, Grok lags the other LLMs in terms of revenues and reach, for the moment. In terms of usage, Anthropic (with Claude), OpenAI (with ChatGPT) and Google (with Gemini) are not only more widely used than Grok, in business setting, but are further along in converting them into revenues. While Grok has been bundled into the X Premium subscriptions, and earned about $80 million in revenues in 2025, it seems to be focusing more on consumers and only on niche portions of the business market. This is the most diffuse and volatile of the three markets, in terms of potential market, since the potential market can run from the tens of billions (if they remain subscription-based) to hundreds of billions or even trillions (if they become replacements for

human labor or massive productivity boosters). It is possible that Grok may concede the larger and more competitive business space to Claude and ChatGPT and focus instead on consumer subscriptions, giving it a smaller market, but one with less competition.

    - *Valuation narrative: The overall LLM market will continue to grow, but more in business applications than in consumer apps, with the market size being determined by how well AI can replicate human labor and regulatory restrictions. xAI will target primarily consumer subscription revenues and niche business applications. That will give it smaller revenues ($80 billion in 2036) than its LLM competitors, but one with less competition and higher margins (50% operating margin), and less reinvestment as it avoids going head-to-head with Anthropic, OpenAI and Google for business use.*

- It is undeniable that SpaceX, as a lead player in three fast-growing and volatile businesses, may be able to use its infrastructure to **expand each of these businesses**. The space launch business, which has generally focused on delivering commercial or government loads and satellites into outer space may become a springboard for space travel, for leisure, research or business. With its satellite broadband offerings, the possibility exists that the technology and the reach will improve to a point where the service can compete with fiber-optic and cable broadband offerings, perhaps at much lower cost. With xAI, the possibility that Grok finds a way to outflank its LLM rivals, including Claude, Gemini and ChatGPT, in terms of business offerings may be low, but it does exist. The recent acquisition of Cursor, a young AI company in the coding space, suggests that xAI has not thrown in the towel on business applications. In truth, these are all options that may not be viable at the moment, but if they become viable, could add immense value.

    - *Valuation narrative: This part of the story is built on the expansion options that SpaceX has to enter large markets, with low probability and high payoff. In a crude attempt to capture this part of the story, I will attach an expected revenue in 2036 to these other businesses of $50 billion and an operating margin of 30%. Since these expansion options, if they do exist, will not show up in the near future, the revenues from these options ramp up after year 6 (2032) in the valuation.*

Pulling these storylines together as valuation inputs, I estimate the following numbers for 2036, for the three business lines and for the expansion options category:

Business	Storyline	Revenues in 2036 ($ millions)	Operating Margin in 2036	Sales to Capital: 2026-30	Sales to Capital: 2031-2035
Launch	Dominant player in a growing launch market; 70% market share of $100 billion market.	$70,000	40%	4.00	2.00
Starlink	Market leader (75%) in satellite broadband, a niche (10%) of very large broadband service market ($1.5 trillion)	$120,000	60%	10.00	5.00
xAI	LLM with focus primarily on consumer subscriptions, rather than business-use	$80,000	45%	2.50	1.50
Expansion Options	Expansion options in space (travel), internet services and AI-related businesses.	$50,000	30%	3.00	3.00

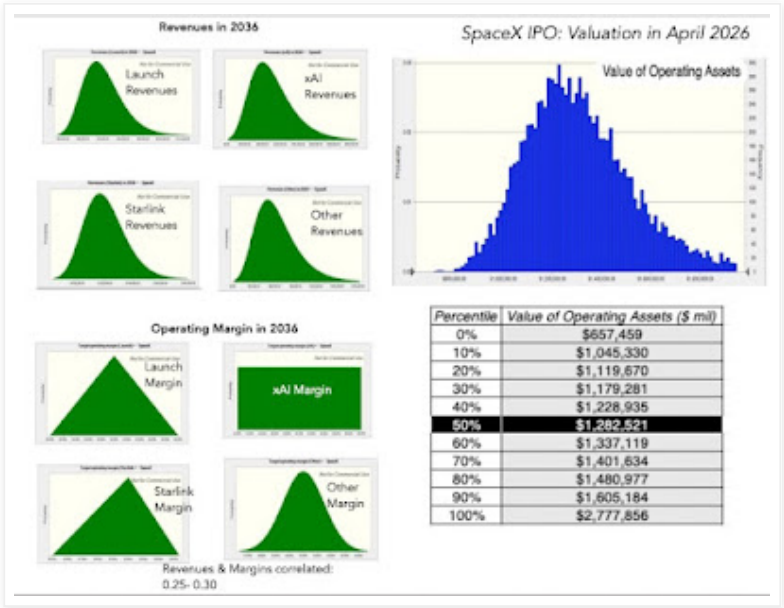
With these numbers in place, and using a cost of capital reflective of SpaceX's business mix (of aerospace/defense, telecom services and AI) of 8%, we can estimate the company's value:

SpaceX						
To Two Trillion and beyond?						Apr-26
SpaceX has three business (for the moment), the space launch, satellite internet and LLM businesses (with xAI). In the first two, it has significant cost advantages and moats that will allow it to maintain dominant market shares, as the businesses grow, in conjunction with high operating margins, driven by favorable unit economics. In the LLM race, Grok (xAI's offering) is more consumer-focused than its competitors (Gemini, Claude and ChatGPT) and is expected to settle into a subscription-based business which is large and profitable.						
The Assumptions						
	Base year	Year 1	Year 10		After year 10	Link to story
Revenues (a)	$ 15,800		4.20%		4.20%	Market leadership and competitive advantages allow for dominant market share.
Operating margin (b	-2.03%	-2.03%	29.27%		48.44%	Solid & improving unit economics in the space launch and satellite internet businesses increase margins.
Tax rate	10.00%	10.00%	25.00%		25.00%	Global tax rate
Reinvestment (c)		Sales to capital (2.66		RIR =	28.00%	Existing investments in infrastructure allow for efficient growth in businesses.
Return on capital	-1.11%	Marginal ROIC =	129.77%		15.00%	Cost of entry will limit competition.
Cost of capital (d)		8.02%	8.00%		8.00%	Weighted business risk.
The Cash Flows						
	Revenues	Operating Margi	EBIT	EBIT (1-t)	Reinvestment	FCFF
1	$38,377	11.89%	$4,563	$4,107	$4,864	-$757
2	$57,916	15.99%	$9,259	$8,333	$7,458	$875
3	$74,865	20.39%	$15,182	$13,664	$6,296	$7,368
4	$89,743	24.59%	$22,063	$19,857	$5,360	$14,491
5	$129,483	29.27%	$37,902	$34,112	$16,523	$17,589
6	$167,587	32.68%	$54,782	$47,640	$31,659	$15,984
7	$205,690	36.42%	$74,902	$62,918	$31,659	$31,259
8	$243,793	40.33%	$98,322	$79,641	$31,659	$47,982
9	$281,897	44.35%	$125,021	$97,516	$31,659	$65,857
10	$320,000	48.44%	$155,000	$116,250	$31,659	$84,591
Terminal year	$333,440	48.44%	$161,510	$121,133	$33,917	$87,215
The Value						
Terminal value			$ 2,295,142			
PV(Terminal value)			$ 1,061,402			
PV (CF over next 10 years)			$ 154,659			
Value of operating assets =			$ 1,216,061			
Adjustment for distress			$ -		Probability of failure =	0.00%
- Debt & Minority Interests			$ -		Not available, since financials for the company are not public. Set to zero in this iteration for cash and debt.	
+ Cash & Other Non-operating assets			$ -			
Value of equity			$ 1,216,061			
- Value of equity options			$ -			
Number of shares			2,416.67		Based upon private company pricing, but may change.	

Download spreadsheet

With my story and inputs, the value that I derive is $1.22 trillion, about 10% below the private market pricing and about a third below the expected IPO pricing, but still astonishingly high for a company with $15.5 billion in revenues in the most recent year, and a host of question marks about corporate governance. Note that in this iteration, I have ignored cash and debt, since I do not have the company's financial statements, but it is unlikely that either will have much of an impact on the value of equity for a company with this high a value for its operating assets.

As you can see, the SpaceX story not only has many moving parts, but is fraught with uncertainty, and without full financials, it does not have a good starting point. That said, as the story plays out, we will get more clarity, and the story will need to be reworked, with the value consequences unclear. For the moment, though, I am uncertain about every input in my SpaceX valuation, but uncertainty is a continuum, and I am less uncertain about some inputs (such as the revenues and margins in the space launch and satellite internet service businesses) than about other inputs (including the revenues and margins of the LLM and expansion businesses). If you are wondering why the cost of capital is only 8% for the company, close to the median cost of capital for a US company, it is because much of the risk here is specific to the company (thus reducing the effect in a diversified portfolio) and cuts in both directions (upside and downside). In fact, if there are outliers, they are more likely to be on the upside than the downside. With these considerations in mind, I tried to be open about how uncertain I feel about my estimates, and the results of a simulation yields the following distribution for value:



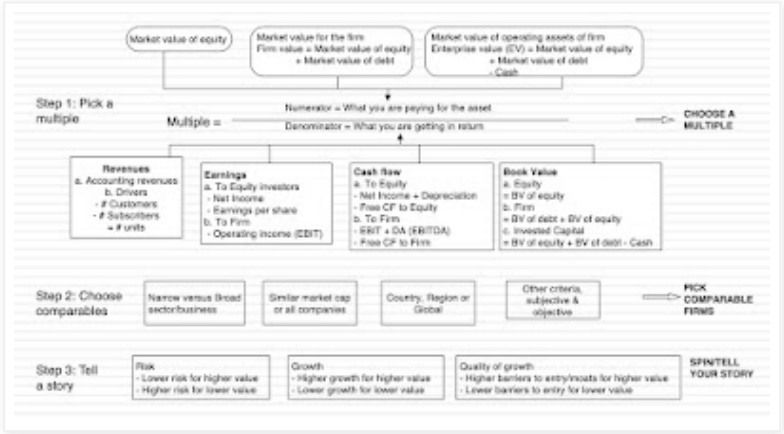
Since the simulation is centered on the same expected values for inputs as I used in my base case, it should come as no surprise that the median value, across ten thousand simulations, of $1.29 trillion is close to the base case valuation of $1.22 trillion. As the pricing for the IPO starts to gain traction, it is worth recognizing that a $1.75 trillion or even a $2 trillion pricing falls in the range of the distribution, though with little or no upside left for an investor paying that price.

If you feel that it is best to wait for the prospectus to be filed, before doing the valuation, I understand but there are three points worth remembering. First tt is unlikely that the prospectus will contain data that will move the intrinsic value story, since none of the numbers in the reported statements will be large enough to alter the immense value coming from expectations of future

growth. Second, while the prospectus will contain estimates of total addressable market and perhaps even profitability, in my experience, it will be hype; expect to see trillions of dollars thrown around nonchalantly for market size. Third, as I see it, it is not an either/or proposition, since I can value the company now, and revisit the valuation when the prospectus comes out, with the advantage being that you are less likely to be swayed by the sales pitch in the prospectus.

Pricing SpaceX

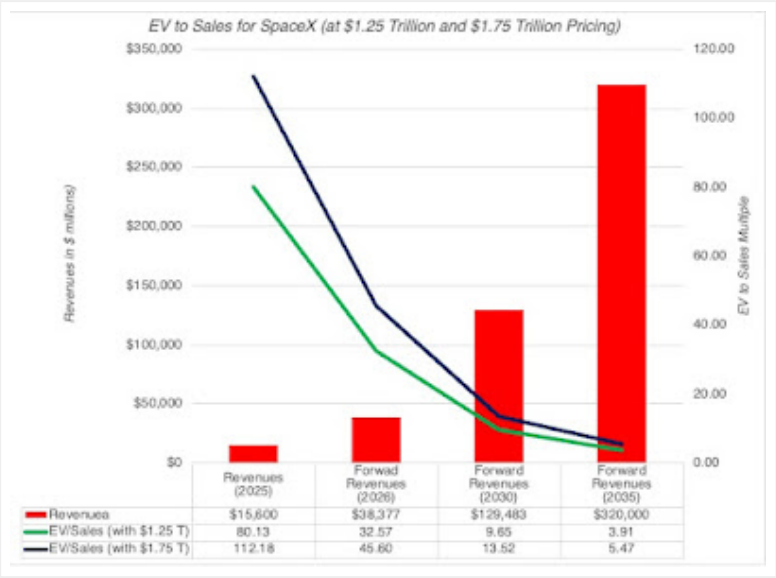
In an initial public offering, companies are priced, not valued, by bankers (for the offering price), by investors (as the stock starts trading) and by observers to make judgments (on whether it over or underpriced). Thus, you can make the argument that the valuation, with all of its moving parts, is irrelevant, and *that you should price SpaceX, not value it,* if your intent is to trade on the IPO. That is a legitimate critique, but the argument that pricing somehow dispenses with the need to make assumptions about market size and profitability or does not have the same uncertainties is not. You can take issue with the intrinsic valuation because of the layers of assumptions that I had to make along the way, and I know that for many investors, either invested already in the company ,or planning to invest in it, a pricing may seem less daunting. While I sympathize, I am afraid that the uncertainty will be just as much of an issue in pricing SpaceX, and to see why, take a look at the steps in the pricing process:



At each step in the process, you will run into issues.

- On the **pricing metric front,** the problem with picking a metric is an information vacuum, with only two scalars, revenues and EBITDA, available, and even if you consider the most recent private market pricing, which priced the company at about $1.25 trillion, as the market value of equity, the absence of debt and cash numbers makes it impossible to back into enterprise value. This problem should be resolved, for the most part, when the company files a full prospectus, but for the moment, if you

assume that the net debt number is close to zero, the resulting enterprise value of $1.25 trillion yields nosebleed multiples of **81 times revenues and 156 times EBITDA for the company, using 2025 numbers**. Even with minimalist information, there will be pricing variants that use expected revenues (or EBITDA) in a future year as a scalar, as can be seen in this graph:



There is no inherent problem with using forward numbers in pricing, as long as you do the same for all of the companies in your peer group, but in the case of SpaceX. it is inevitable that bullish analysts, unable to justify the sky high pricing values with trailing 12-month numbers, will resort to using forward pricing, and add to the bias, by inflating revenues in future years. In the graph, I have used the forecasted revenues in 2030 and 2035, from my intrinsic valuation, and the EV to Sales ratio drops from 80.13 (112.18) to 3.91 (5.47), using the private company (estimatedIPO offering) pricing of $1.25 trillion ($1.75 trillion) as the enterprise value.

- The even bigger challenge in pricing SpaceX will be in the second step, where **you have to find comparable firms (or a peer group) to base your pricing on**. There is obviously no company out there that is remotely similar to SpaceX, as a composite company, and even if you break it down into businesses, it is likely that you will hit roadblocks. On the space launch business, using publicly traded aerospace and defense companies like Boeing and Northrop Grumman is a non-starter, because they are low growth, lower-margin businesses, unlike SpaceX. Palantir may seem like a logical alternative, and while it may have high growth potential, it is a data/software company that does not have the infrastructure needs that SpaceX does. On the

internet service business again, there are conventional telecom firms like Verizon and T-Mobile, but the economics of their offerings are different, and they are not growth companies. In April 2026, I computed the multiples of revenues and EBITDA that publicly traded companies in the aerospace/defense, internet services businesses and technology companies trade at, and they are far lower than what SpaceX can be expected to trade at, if it goes public at $1.5 to $2 trillion:

	EV to Sales		EV to EBITDA	
	Aerospace & Defense	*Internet Services*	*Aerospace & Defense*	*Internet Services*
First Decile	1.33	0.42	11.08	5.38
First Quartile	2.37	0.93	17.03	8.02
Median	5.67	2.10	28.58	21.22
Third Quartile	12.95	8.40	54.45	36.22
Ninth Decile	33.74	22.52	117.40	129.15

Finally, on xAI, there are other LLMs, almost all of which are private companies now, and while you can use the pricing from most recent VC rounds, you are on shaky grounds. Even if you forged ahead with a peer group of high-growth, tech companies, your pricing will almost certainly have to revolved around revenues, rather than profits, and based upon very small samples.

- The one part of pricing that you will almost certainly see **is the story telling**, especially with analysts who are locked into finding SpaceX to be a buy, with the pricing more of an ex-post rationalization than a analytical tool. No matter what peer group you pick, SpaceX will be priced higher than comparable firms, and to back the argument that it is still a good investment, you will hear stories of its large potential market, significant competitive advantages and profitability. All of these stories are grounded in truth, but they are empty if they remain stories. I will predict that there will be far more buy than sell or hold recommendations for SpaceX, when it does go public, with analysts doing pricing gymnastics with forward multiples, hand-picked peer groups and fairy tales to justifying their positions.

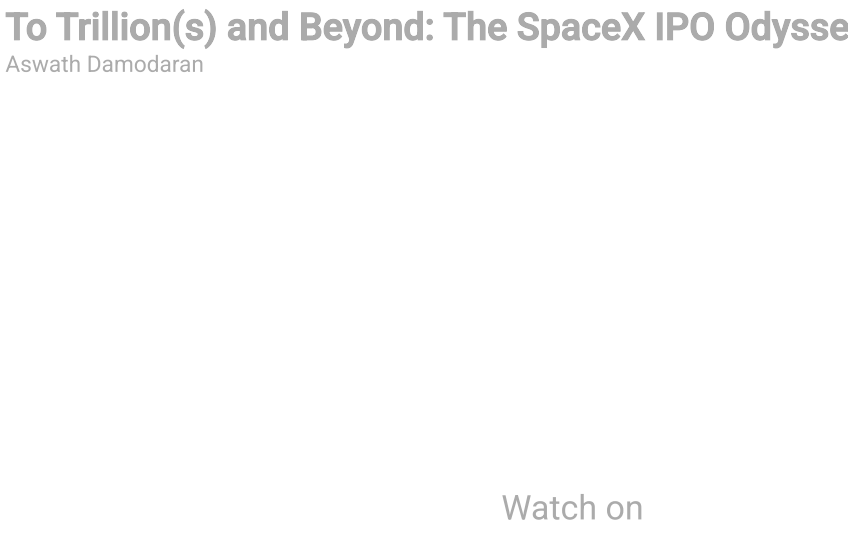
Pricing is an exercise in data analysis, and any pricing of SpaceX will reflect the statistical limitations of the data. Put simply, the only difference between intrinsic valuation and pricing, when it comes to the uncertainties you face with SpaceX, is that with the former (intrinsic valuation), you have to face up to the uncertainties and make your best explicit judgments (on revenues, margins and reinvestment), whereas with the latter (pricing), they remain implicit. If bias is your biggest adversary in assessing SpaceX, and you use pricing, it is very likely that you will find a pricing metric and hand-picked peer group to reflect your biases, and then tell yourself a story on why SpaceX is cheap or expensive.

The Bottom Line

SpaceX is an engineering marvel that has shown its naysayers, which included almost every luminary in the space community, to be wrong. That said, for potential investors, there are lessons to be learned from watching Musk's stewardship of Tesla. As with all of Musk's creations,

SpaceX will be a shape shifting entity, frustrating investors who expect companies to follow linear paths in the corporate life cycle, going from young growth to maturity; it will shift from one narrative to another, often with no advance warning, causing whiplash for investors. When I bought Tesla in 2019, after its stock had taken a beating, I described the company as my corporate teenager, and with Musk in full control, SpaceX is likely to follow the same unpredictable path. That makes it a difficult company to buy, but it makes it an even more dangerous company to sell short, as Tesla short sellers have discovered in the last two decades. With all that said, SpaceX is a unique company with immense competitive advantages, and while I would not be interested in buying at the rumored IPO pricing of $1.75 trillion, it is one big correction away from being fairly priced or even cheap. If that happens, I will be a buyer, but will do so with the recognition that this company comes packaged with a founder who is both uniquely gifted and deeply flawed, and complaining about the parts of Musk you do not like, while enjoying the fruits of the aspects that you do, is unfair.

YouTube Video



Spreadsheets

1. Valuation of SpaceX in April 2026 (Pre-IPO, and with limited financials)

Posted by Aswath Damodaran at 5:20 PM No comments:
Labels: IPO, Optionality, Value of growth, Valuing Young companies

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